

Chiral Boundary Mode Ensembles in Dynamical Chern Insulators

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Measurements in an adaptive quantum circuit generate both a conditional quantum state and a classical record drawn from its Born probability. We study the interplay of these two objects in a finite-range free-fermion circuit whose bulk realizes class-A Chern dynamics and whose local controller programs a one-dimensional domain wall. The conditional wall states form **[STATE-CFT RESULT]**, while their propagation is probed in three complementary ways: orientation-resolved modular flow, the full Born linear response including record reweighting, and fixed-record flux insertion. We further extract **[RECORD RESULT]** from defect contributions to the Born self-information and a charge-resolved replicated transfer spectrum. Exact occupation projectors and record-conditioned gain or loss render the usual invertible one-body transfer matrix singular. We resolve this obstruction by differentiating the normalized covariance map, obtaining a semi-invertible tangent cocycle whose boundary-localized low-exponent sector yields **[TANGENT-PURIFICATION RESULT]**. The resulting framework distinguishes the universal physics of the physical trajectory ensemble from both a selected non-Hermitian branch and the generally mixed unconditional channel, and makes a programmable topological interface an operational probe of monitored quantum dynamics.

I. INTRODUCTION

Measurements do more than decohere a quantum system. When their outcomes are retained, they generate record-conditioned trajectories whose entanglement and information content can undergo sharp transitions even when ordinary averaged observables remain smooth. The resulting measurement-induced phases have connected quantum information, nonequilibrium statistical mechanics, and many-body dynamics through problems of entanglement, purification, error correction, and information flow [1–7]. A defining feature of this setting is that the observed record is not incidental: through the Born rule, it supplies the probability measure on the physical ensemble of quantum trajectories.

Topology makes the role of this ensemble especially sharp. In equilibrium, a mismatch of bulk invariants is exposed at an interface through protected boundary degrees of freedom; for a Chern insulator, the boundary supports a chiral complex fermion [8, 9]. Periodic driving extends bulk–boundary correspondence from a Hamiltonian to an evolution law, while monitored circuits admit symmetry-enriched and topologically ordered trajectory phases with no reduction to a single ground-state problem [10–13]. In one spatial dimension, measurement-only spin circuits can further support symmetry-enriched percolation and gapless SPT steady-state ensembles with protected endpoint modes diagnosed through purification and symmetry-flux string observables [14]. Viewed this way, a monitored phase is a dynamical phase in a causally generated temporal-disorder ensemble: each outcome selects the map applied at that time, while the Born rule and adaptive feedback endow the sequence with an intrinsic physical measure and make later maps depend on earlier outcomes. The relevant robustness question is therefore not only whether topology survives spatial dis-

order, but whether a trajectory-resolved bulk–boundary correspondence persists under this physical temporal disorder. This raises a direct dynamical question: when two monitored phases meet, what replaces the equilibrium boundary theory in the physical trajectory ensemble?

For Gaussian fermions, nonunitary circuit evolution can be related to single-particle transfer and localization problems in the Altland–Zirnbauer classes, providing a natural language for monitored topology and criticality [15, 16]. Recent work has constructed programmable defects in monitored free-fermion dynamics and identified topological boundary sectors through Lyapunov spectra [17–19]. These developments establish that topology can be assigned to monitored evolution without first replacing it by an equilibrium Hamiltonian. They also expose several inequivalent notions of an edge: a mode of a selected transfer problem, a Lyapunov sector of a stochastic product, and a critical state carried by typical physical trajectories. The immediate starting point for the present work is the adaptive Chern circuit of Bhuiyan, Pan, and Jian, which realizes a postselection-free class-A trajectory ensemble in two dimensions, permits a local domain wall to be programmed into the controller, and exhibits anomalously slow purification at that wall [20]. What remained unresolved was the theory living on the interface: whether the wall ensemble is critical and chiral, how its slow dynamics survives exact projective rank loss, and whether trajectory-resolved measurement statistics carry universal defect data.

Monitored free fermions display a particularly rich—and subtle—critical landscape. Continuous and projective monitoring can produce long finite-scale regimes with logarithmic or stronger subvolume entanglement, localization transitions, conserved-charge dynamics, and nonlinear-sigma-model descriptions [21–25]. Above one spatial dimension, related theories support genuine measurement-

induced transitions, whereas increasingly large studies of short-range one-dimensional models show that an apparent extended critical regime can cross over only at parametrically large scales [26, 27]. A logarithmic entropy fit is therefore not sufficient evidence for the wall considered here. The same boundary sector must appear consistently in entanglement, charge fluctuations, real-space localization, purification, and orientation-sensitive propagation.

Adaptive circuits provide another nearby perspective. Measurements, refreshed ancillas, and feedforward can prepare and stabilize entangled targets and can, in suitable constructions, accelerate or reduce the depth of preparation of a common wave function or ground-state sector [28–31]. In that setting, the measurement record often labels a correctable byproduct, and classical coordination can be used to make different branches agree on the desired output. Interactive dynamics can instead support trajectory-entanglement, steering, control, and absorbing transitions, whose conditioned and averaged critical behavior need not coincide [32–34]; related frustration-free protocols emphasize transport-limited convergence [35]. Our protocol lies between these settings. The correction is local and every measurement outcome is retained, but it does not erase the record or collapse all outcomes onto one wave function; repeated feedback stabilizes an ensemble of inequivalent, charge-fluctuating states that share the same long-distance wall physics. The wall is consequently a steady defect of the stochastic dynamics, rather than a compiled representative of a predetermined conformal ground state. The finite-support controller admits a finite-color schedule; Sec. II relates its physical depth to the adaptive preparation bound.

Microscopically, the controller is shaped by the real-space obstruction of a Chern band. Exponentially localized orthonormal Wannier functions are unavailable, whereas localized nonorthogonal overcomplete frames provide suitable measurement modes; strictly local Gaussian tensor-network representatives face related gaplessness and parent-Hamiltonian obstructions [36–38]. We use finite-support approximations to these frames, measure their occupations, and correct an undesired outcome by exchanging a fermion with a freshly prepared occupied or empty ancilla. Nonorthogonal monitored orbitals can themselves generate anomalous information transport [39]; here their finite-support geometry instead implements a topological controller whose wall is tested against matched trivial circuits. This ideal perfect-correction step is the fresh-ancilla realization of the earlier adaptive circuit, not an additional stochastic correction probability. The same local update creates a domain wall either by joining distinct controller frames or by terminating one frame at a chosen interface. Agreement between these constructions is important because it separates universal wall physics from the microscopic manner in which the control rule is patched. Native-fermion processors provide several of the required primitives, although a full implementation of the present cycle remains a future objective [40].

The physical question requires four related objects to be kept distinct: a selected conditional Kraus word, the probability-weighted ensemble of conditional states, the classical distribution of measurement records, and the unconditional state obtained after discarding those records. Section II defines all four after constructing the microscopic circuit. Non-Hermitian Hamiltonians and monitored implementations give powerful descriptions of selected or effective branches, but a branch does not determine the probability-weighted ensemble [41, 42]. Indeed, Born-rule and forced-measurement free-fermion circuits can realize different universal behavior even within the same single-particle symmetry class [43, 44]. Probabilistic filters can implement ground-state filtering in suitable settings [45]. Our focus is instead the complete Born ensemble as the physical dynamical object.

The unconditional channel provides a complementary, but different, description. Engineered dissipation can prepare topological steady states and boundary structures [46–48], while finite-range quadratic Lindblad dynamics in more than one dimension cannot converge at a nonvanishing size-independent rate to a unique pure Gaussian topological state under the assumptions of the dissipative preparation obstruction [49]. Our construction respects this obstruction in two complementary ways. With their full exponential tails, the exact OW stabilizers are local but not finite range, which is an allowed route to a common pure Chern target. At finite n_{shell} the controller is strictly finite range, but truncation destroys the common pure Chern dark state: the conditioned records produce inequivalent trajectories and the discarded-record state $\bar{\rho}_T$ is generally mixed. A critical wall may additionally carry a relaxation scale that closes with system size. Thus neither overcompleteness, fresh ancillas, nor adaptive feedback alone is being invoked as a loophole. The theorem constrains the unconditional state, whereas the normalized conditional trajectories follow a different evolution; different unravelings of the same averaged evolution can retain different trajectory statistics [50]. We therefore compare conditional and unconditional observables directly instead of assigning the trajectory CFT to the averaged channel.

The record itself supplies a third level of description. Quantum-jump trajectories admit large-deviation and thermodynamic ensembles, while continuous-measurement and fermion-counting theories relate observed stochastic currents to tilted full-counting-statistics generators [51–54]. More directly, Born-weighted record transfer matrices can encode effective central charges and operator dimensions at measurement-induced critical points [55]. We ask whether this construction admits a defect version tied to a programmable class-A interface. A conformal interpretation requires a replicated transfer problem, subtraction of its extensive bulk and boundary terms, and consistent universal sector data from independent finite-size observables. This construction yields **[RECORD-ONLY BOUNDARY DATA]** without reconstructing every conditional state.

We first characterize the state ensemble at the programmed wall. The numerical observables combine chord-length scaling of the wall excess entropy, quantum interval-charge fluctuations, density correlations, and real-space entanglement contours. The charge variance is evaluated within each trajectory before the Born average, so its logarithmic coefficient probes the $U(1)$ current algebra rather than the sample-to-sample wandering of the mean charge. Together these observables determine **[STATE-CFT DATA]** and localize the critical contribution at the interface. Starting from a maximally mixed state, the same wall produces **[PURIFICATION RESULT]**, in contrast with the gapped bulk dynamics. Because state-level conformal data determine $c_L + c_R$ rather than the propagation direction, we test chirality separately through modular charge drift, the full momentum- and frequency-resolved Born response, and fixed-record flux insertion. The paired wall orientations provide an internal reversal test for all three probes, while wall-off circuits distinguish directed propagation from a schedule-dependent drift.

We then develop a transfer description adapted to exact projectors. Differentiating the normalized Gaussian update produces a semi-invertible tangent cocycle that remains well defined when an occupation measurement removes rank. Its spectrum exhibits **[TANGENT-GAP RESULT]**, its slow subspace is **[LOCALIZATION RESULT]**, and its low-exponent density is compared directly with the purification law. In parallel, we use replicated record weights and charge-resolved transfer sectors to extract **[RECORD-CFT RESULT]**. Comparing both constructions with the deterministic averaged covariance map isolates which singular behavior belongs to conditioned trajectories and which survives after the record is discarded.

Taken together, the state, tangent, and record descriptions characterize different aspects of the same programmed interface. The circuit samples a Born-weighted, charge-fluctuating ensemble, while **[SHARED CHIRAL-WALL CONCLUSION]** determines the universal structure common to its trajectories. This distinction is also operational. A quantum simulator samples the ensemble by retaining every physical record, while the classical record offers a potential route to universal information that can be compared directly with state-tomographic observables. Figure 1 introduces the adaptive instrument and the two domain-wall constructions. The following sections then establish the wall state, its orientation, the record theory, the tangent mechanism, and the distinction from unconditional evolution.

II. ADAPTIVE CIRCUIT AND PROGRAMMED DOMAIN WALLS

We now specify the adaptive circuit. We follow the notation and construction sequence of BPJ: first the Chern parent and its projected OW modes, then their occupation stabilizers, and finally the measurement-and-feedforward

cycle. The only band-structure convention change is $\alpha_{\text{here}} = -\alpha_{\text{BPJ}}$, which keeps the numerical input $\alpha_{\text{top}} = 1$ associated with a lower band of Chern number $+1$. Figure 1 summarizes the construction.

A. Chern parent, OW modes, and stabilizers

Consider a square lattice of $N_x \times N_y$ unit cells with two complex fermions $\hat{c}_{\mathbf{r},1}$ and $\hat{c}_{\mathbf{r},2}$ per cell. We write $\hat{c}(\mathbf{k}) = (\hat{c}_1(\mathbf{k}), \hat{c}_2(\mathbf{k}))^\top$ and define

$$\begin{aligned}\hat{\mathcal{H}}_{\text{CI}}(\alpha) &= \sum_{\mathbf{k}} \hat{c}^\dagger(\mathbf{k}) [\vec{n}_\alpha(\mathbf{k}) \cdot \vec{\sigma}] \hat{c}(\mathbf{k}), \\ \vec{n}_\alpha(\mathbf{k}) &= (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \\ P_\pm(\mathbf{k}; \alpha) &= \frac{1}{2} \left[\mathbf{1} \pm \frac{\vec{n}_\alpha(\mathbf{k})}{|\vec{n}_\alpha(\mathbf{k})|} \cdot \vec{\sigma} \right].\end{aligned}\quad (1)$$

Here P_+ and P_- project onto the upper and lower single-particle bands, respectively. With

$$\mathcal{C}[P] = \frac{1}{2\pi i} \int_{\text{BZ}} d^2k \text{Tr}(P[\partial_{k_x} P, \partial_{k_y} P]), \quad (2)$$

the occupied lower band has

$$\mathcal{C}[P_-] = \begin{cases} -1, & -2 < \alpha < 0, \\ +1, & 0 < \alpha < 2, \\ 0, & |\alpha| > 2. \end{cases} \quad (3)$$

Thus $\alpha_{\text{top}} = 1$ gives $\mathcal{C} = +1$, while $\alpha_{\text{triv}} = 30$ is trivial. Equations (1) and (3) are BPJ's equations after $\alpha_{\text{BPJ}} \mapsto -\alpha$; this convention is fixed throughout the state, flux, and chirality analyses.

Following BPJ, choose the cell-local trial spinors

$$\tau_A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \tau_B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (4)$$

and project them into either band:

$$|\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle = P_\pm(\alpha)|\mathbf{r}, \tau_\nu\rangle, \quad \nu = A, B. \quad (5)$$

The corresponding normalized localized fermion operator is

$$\begin{aligned}\hat{\chi}_{\mathbf{r},\nu,\pm}^{(\infty)} &= \sum_{\mathbf{r}',\mu'} W_{\mathbf{r},\nu,\pm}^{(\infty)}(\mathbf{r}', \mu') \hat{c}_{\mathbf{r}',\mu'} \\ &\propto \frac{1}{\sqrt{N_x N_y}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \tau_\nu^\dagger P_\pm(\mathbf{k}; \alpha) \hat{c}(\mathbf{k}),\end{aligned} \quad (6)$$

where $W_{\mathbf{r},\nu,\pm}^{(\infty)}(\mathbf{r}', \mu') = \langle \widetilde{W}_{\mathbf{r},\nu,\pm} | \mathbf{r}', \mu' \rangle / \|\widetilde{W}_{\mathbf{r},\nu,\pm}\|$. These OW modes are localized but not mutually orthogonal. Their unnormalized states obey the tight-frame identity

$$\sum_{\mathbf{r},\nu=A,B} |\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle \langle \widetilde{W}_{\mathbf{r},\nu,\pm}| = P_\pm, \quad (7)$$

so the two trial families together span the corresponding band.

FIGURE 1 PLACEHOLDER

- (a) Flattened two-band parent $\hat{\mathcal{H}}_{\text{CI}}$, lower/upper targets, and A/B projected OW modes.
- (b) Truncate-then-renormalize construction for the four 3×3 controller modes; inset: the displaced overlap zero.
- (c) Born occupation measurement, target-dependent fresh ancilla, deterministic fSWAP, and ancilla trace/reset.
- (d) One complete cycle: the BPJ-ordered $A-, A+, B-, B+$ word at every active center, together with the four ensemble objects.
- (e) Explicit-interface and support-terminated walls, their bond locations, and the orientation convention.
- (f) Bulk Chern-marker/twist convergence and matched comparison of the two wall constructions.

FIG. 1. Adaptive Chern controller and programmed domain walls. (a) The lower band of the flattened parent at $\alpha_{\text{top}} = 1$ has $\mathcal{C} = +1$ and is targeted as occupied, whereas the upper band is targeted as empty. Our mass parameter has the opposite sign from BPJ. (b) Each projected mode is multiplied by the real-space window and then normalized independently; $n_{\text{shell}} = 1$ has 3×3 -cell support. (c) The occupation outcome is Born random. Only a mismatch invokes an occupied or empty fresh ancilla and a deterministic fSWAP; the outgoing ancilla is then discarded without conditioning. (d) The overlapping channels are visited in the BPJ sequence, lower then upper for A followed by B , and form a causal ordered word rather than four commuting stabilizer layers. (e) The interface may be programmed by patching two local controller frames or by terminating the topological frame. (f) Topology and wall universality are checked independently.

Define the OW occupation operators

$$\hat{\mathcal{N}}_{\mathbf{r},\nu,\pm}^{(\infty)} = \hat{\chi}_{\mathbf{r},\nu,\pm}^{(\infty)\dagger} \hat{\chi}_{\mathbf{r},\nu,\pm}^{(\infty)}. \quad (8)$$

The filled-lower-band state $|\text{CI}\rangle\rangle$ is uniquely fixed by the BPJ stabilizing conditions

$$\hat{\mathcal{N}}_{\mathbf{r},\nu,-}^{(\infty)} |\text{CI}\rangle\rangle = |\text{CI}\rangle\rangle, \quad (\mathbf{1} - \hat{\mathcal{N}}_{\mathbf{r},\nu,+}^{(\infty)}) |\text{CI}\rangle\rangle = |\text{CI}\rangle\rangle \quad (9)$$

for every $\mathbf{r}$ and $\nu = A, B$. The lower-band modes therefore target occupation one, whereas the upper-band modes target occupation zero.

B. Finite support and programmed walls

The physical circuit uses compact approximations to these OW modes. Let $d_{N_\mu}(r_\mu, r'_\mu)$ denote the shortest periodic displacement and, for $n = n_{\text{shell}}$, define the square window

$$M_{\mathbf{r}}^{(n)}(\mathbf{r}') = \mathbf{1}[|d_{N_x}(x, x')| \leq n, |d_{N_y}(y, y')| \leq n]. \quad (10)$$

The windowed wavefunction and its operator are

$$W_{\mathbf{r},\nu,\pm}^{(n)}(\mathbf{r}', \mu') = \frac{M_{\mathbf{r}}^{(n)}(\mathbf{r}') \widetilde{W}_{\mathbf{r},\nu,\pm}(\mathbf{r}', \mu')}{\|M_{\mathbf{r}}^{(n)} \widetilde{W}_{\mathbf{r},\nu,\pm}\|}, \quad (11)$$

$$\hat{\chi}_{\mathbf{r},\nu,\pm}^{(n)} = \sum_{\mathbf{r}', \mu'} W_{\mathbf{r},\nu,\pm}^{(n)}(\mathbf{r}', \mu') \hat{c}_{\mathbf{r}', \mu'}.$$

Thus the mode is first truncated and only then renormalized. Before a wall mask is imposed, $n_{\text{shell}} = 1$ has exactly 3×3 cells of support, or at most 18 microscopic orbitals. At $\alpha = 1$, it retains 0.992155 of the untruncated squared norm and a target-band overlap zero with winding $\nu_{\text{ov}} = -1$, displaced from $\mathbf{k}/\pi = (0.5, 0)$ to approximately

$(0.4922, 0)$. In our convention $\mathcal{C}[P_-] = -\nu_{\text{ov}} = +1$. This is an overlap-obstruction diagnostic, not a Chern number of the compact spinor, which defines a smooth global section and has $\mathcal{C}_{\text{compact}} = 0$. Appendix B gives the convolution and winding analysis. Because windowing introduces out-of-band weight, the compact occupations no longer share the exact common eigenstate in Eq. (9); the finite-range circuit stabilizes an ensemble rather than one exact parent-Hamiltonian ground state. Consequently, the finite-shell cutoff is an approximation at the level of the local controller, not a compactly supported representative of the Chern bundle. Its large retained norm and nonzero overlap winding preserve the imprint of the target projector, but not the exactly flat topological dark subspace excluded for a finite-range dissipative preparation with a unique pure steady state and nonvanishing thermodynamic relaxation rate [49].

We program the wall in two ways. Let $\Theta_{\text{top}}(\mathbf{r}) = \mathbf{1}[x_L \leq x \leq x_R]$ and

$$\alpha_{\mathbf{r}} = \begin{cases} \alpha_{\text{top}}, & \Theta_{\text{top}}(\mathbf{r}) = 1, \\ \alpha_{\text{triv}}, & \Theta_{\text{top}}(\mathbf{r}) = 0. \end{cases} \quad (12)$$

For the *explicit interface*, every center remains active and Eq. (11) is formed using the uniform-parent projector $P_{\pm}(\mathbf{k}; \alpha_{\mathbf{r}})$ assigned to that center. Compact tails may cross the interface. This construction patches locally uniform controller frames; it is not the spectral projector of one inhomogeneous Hamiltonian.

For the *support-terminated wall*, define

$$D_{\mathbf{r}}(\mathbf{r}') = \Theta_{\text{top}}(\mathbf{r})\Theta_{\text{top}}(\mathbf{r}') + [1 - \Theta_{\text{top}}(\mathbf{r})][1 - \Theta_{\text{top}}(\mathbf{r}')] \quad (13)$$

and, for active centers $\mathbf{r}$ in the topological slab, use

$$W_{\mathbf{r},\nu,\pm}^{(n,\text{term})} = \frac{D_{\mathbf{r}} M_{\mathbf{r}}^{(n)} \widetilde{W}_{\mathbf{r},\nu,\pm}(\alpha_{\text{top}})}{\|D_{\mathbf{r}} M_{\mathbf{r}}^{(n)} \widetilde{W}_{\mathbf{r},\nu,\pm}(\alpha_{\text{top}})\|}. \quad (14)$$

The diagonal masks commute and are both imposed before one normalization; the order written in Eq. (14) has no physical content. Only topological-slab centers are subsequently updated. The exterior is prepared as a Born-sampled onsite product state and is excluded from the OW word. We denote its stored preparation record by ζ , its probability by p_ζ , and the resulting initial state by $\rho_{0|\zeta}$. For the explicit-interface circuit, ζ is empty and $p_\zeta = 1$. Periodicity in x places oppositely oriented walls on the bonds $x_L - \frac{1}{2}$ and $x_R + \frac{1}{2}$. The globally diagonalized inhomogeneous exact- H problem is a third, deterministic benchmark and is neither circuit construction. We find **[AGREEMENT OF THE TWO WALL CONSTRUCTIONS]** after transverse convergence [Fig. 1(e,f)].

C. Perfect correction and one complete BPJ-ordered cycle

We now define the stochastic update. For a normalized controller mode $\hat{\chi}_j = \sum_i \chi_{j,i} c_i$, where $j = (\mathbf{r}, \nu, \sigma)$ and $\sigma = \pm$, let $\hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j$, $\Pi_{j,1} = \hat{N}_j$, and $\Pi_{j,0} = \mathbf{1} - \hat{N}_j$. A Gaussian state is represented by

$$C_{ij} = \text{Tr}(\rho c_i^\dagger c_j), \quad G = 2C - \mathbf{1},$$

$$p_{j,m}(G) = \frac{1 + \eta_m \chi_j^\dagger G \chi_j}{2}, \quad \eta_m = 2m - 1. \quad (15)$$

The target occupations are $s_- = 1$ and $s_+ = 0$. Conditional on the Born outcome m , perfect correction uses

$$\begin{array}{c|cc} & m=0 & m=1 \\ \hline s=0 & K_{0|0} = \Pi_0 & K_{1|0} = \hat{\chi} \\ s=1 & K_{0|1} = \hat{\chi}^\dagger & K_{1|1} = \Pi_1 \end{array}, \quad \rho'_{m|s} = \frac{K_{m|s} \rho K_{m|s}^\dagger}{p_m}. \quad (16)$$

The channel index j is suppressed in the table. The identities $K_{m|s}^\dagger K_{m|s} = \Pi_m$, $\Pi_s K_{m|s} = K_{m|s}$, and $\sum_m K_{m|s}^\dagger K_{m|s} = \mathbf{1}$ show that the correction preserves the Born probability and resets the measured OW mode to its target occupation.

This ideal model is obtained from a local, number-conserving dilation of the BPJ correction step. On a mismatch, prepare a fresh ancillary mode $\hat{d}$ in $|s\rangle_d$, apply

$$\text{fSWAP}(\hat{\chi}, \hat{d}) = \exp\left(i\frac{\pi}{2} \hat{X}^\dagger \hat{X}\right), \quad \hat{X} = \hat{\chi} - \hat{d},$$

$$|m\rangle_\chi |s\rangle_d \mapsto |s\rangle_\chi |m\rangle_d. \quad (17)$$

and discard or reset the outgoing factorized ancilla. No favorable outcome is selected. Total charge is conserved on the enlarged physical-ancilla system even though the reduced physical trajectory contains gain and loss. Unlike BPJ's reusable ancillary layer, the idealization integrates out a fresh target-state ancilla after every mismatch and therefore has no separate bath-redistribution stage.

For later use, write $P_j = \chi_j \chi_j^\dagger$, $Q_j = \mathbf{1} - P_j$. The exact conditional covariance map is

$$\Phi_{m|s}^{(j)}(G) = Q_j (\mathbf{1} + \eta_m G P_j)^{-1} G Q_j + \eta_s P_j. \quad (18)$$

The inverse is a rank-one resolvent and exists on every nonzero-probability branch. Equation (18), rather than an unnormalized non-Hermitian propagator, is differentiated in Sec. VI.

Algorithm 1: One perfect-correction cycle.

1. Choose the ordered active centers $\pi_t = (\mathbf{r}_{t,1}, \dots, \mathbf{r}_{t,N_A})$.
2. For each center, visit $\nu = A, B$. First measure $\hat{N}_{\mathbf{r},\nu,-}$; if $m_- = 0$, supply an occupied fresh ancilla and apply fSWAP.
3. Next measure $\hat{N}_{\mathbf{r},\nu,+}$; if $m_+ = 1$, supply an empty fresh ancilla and apply fSWAP.
4. Update G with Eq. (18) after every outcome, and store the complete causal record together with π_t .

Thus the within-center word is exactly the BPJ sequence

$$\mathcal{Q}_{\text{BPJ}} = ((A, -), (A, +), (B, -), (B, +)),$$

$$(s_q)_{q \in \mathcal{Q}_{\text{BPJ}}} = (1, 0, 1, 0). \quad (19)$$

Let $j = 1, \dots, D_T$ enumerate the resulting global chronological word. For prescribed center words $\pi = (\pi_1, \dots, \pi_T)$, exterior record ζ , and OW outcomes ξ , the complete operator and causal probability are

$$K_{\xi|\pi} = \overleftarrow{\prod_{t=1}^T} \overleftarrow{\prod_{\ell=1}^{N_A}} \overleftarrow{\prod_{q \in \mathcal{Q}_{\text{BPJ}}}} K_{m_{t\ell q}|s_q}^{(\mathbf{r}_{t,\ell}, q)},$$

$$p_{\xi|\pi, \zeta} = \text{Tr}(K_{\xi|\pi} \rho_{0|\zeta} K_{\xi|\pi}^\dagger) = \prod_{j=1}^{D_T} \frac{1 + \eta_{m_j} \chi_j^\dagger G_{<j|\zeta} \chi_j}{2}. \quad (20)$$

The arrows mean that later operations act on the left, and each Born factor is evaluated on the covariance produced by all preceding operations.

The four ensemble objects introduced in Sec. I are

$$\rho_{\zeta, \xi|\pi} = \frac{K_{\xi|\pi} \rho_{0|\zeta} K_{\xi|\pi}^\dagger}{p_{\xi|\pi, \zeta}},$$

$$\bar{\rho}_{T|\pi} = \sum_{\zeta, \xi} p_\zeta p_{\xi|\pi, \zeta} \rho_{\zeta, \xi|\pi},$$

$$\mathbb{P}(\pi, \zeta, \xi) = P_{\text{sch}}(\pi) p_\zeta p_{\xi|\pi, \zeta}, \quad \bar{\rho}_T = \sum_{\pi} P_{\text{sch}}(\pi) \bar{\rho}_{T|\pi}. \quad (21)$$

Thus $K_{\xi|\pi}$ is a selected conditional OW word; $\{p_\zeta p_{\xi|\pi, \zeta}, \rho_{\zeta, \xi|\pi}\}$ is the physical Born ensemble at fixed schedule; $p_\zeta p_{\xi|\pi, \zeta}$ is its classical record law; and $\bar{\rho}_{T|\pi}$ is the

discarded-record state. The physical outcomes (ζ, ξ) constitute causal temporal disorder conditional on π . If the center ordering is randomized, P_{sch} describes additional externally chosen disorder and must be stored separately. Because neighboring OW projectors overlap, their order is physical: randomized center words are the production choice, while raster and reverse-raster center words are controls.

The controller modes are derived from flattened uniform-parent projectors, but truncation, chronological measurement, Born sampling, and feedback do not equal evolution under the exact domain-wall Hamiltonian. Dashed exact- H curves use its ground-state covariance only as a matched finite geometry benchmark. Real-space markers and twisted-boundary estimators give **[CHERN/BOTT CONVERGENCE AND $n_{\text{shell}} = 1$ ERROR]** [Fig. 1(f)] [56–58]. Appendix D gives the finite-color compilation and the adaptive depth bound. For stationary observables, every production trajectory undergoes its own size-dependent convergence interval before its observation window is collected, and already Born-sampled trajectories enter ensemble averages with equal Monte Carlo weight.

III. CRITICAL STATE ENSEMBLE AT THE WALL

We first characterize the steady-state wall independently of its direction of propagation. Figure 2 combines interval entanglement, $U(1)$ charge fluctuations, real-space contours, correlations, and purification to test whether the Born trajectories share a one-dimensional critical boundary theory.

For a wall interval of length ℓ , we use the chord distance

$$d_L(\ell) = \frac{L}{\pi} \sin\left(\frac{\pi\ell}{L}\right), \quad (22)$$

and define the wall-excess entropy by subtracting a matched bulk or wall-off geometry before fitting

$$\Delta S(\ell) = a_S \log d_L(\ell) + b_S + \delta_S(\ell, L). \quad (23)$$

The conversion between a_S and the central charge is fixed by whether the chosen cut contains one wall branch, two oppositely oriented walls, or a boundary interval. We use that geometry without changing the convention between the exact- H benchmark and the Born ensemble. Figure 2(a) gives **[a_S , c_{edge} , AND LARGE- L DRIFT]**. This entropy scaling determines the total critical content seen by the cut, $c_L + c_R$, but not its orientation [59, 60].

Charge cumulants supply independent state data for free fermions [61, 62]. For the $U(1)$ boundary theory, the current-algebra normalization derived in Appendix F relates the logarithmic interval variance to its current level. With $N_A = \sum_{j \in A} n_j$, we first form the quantum variance inside each trajectory and only then average over

trajectory-resolved records,

$$F_{A,\xi} = \langle N_A^2 \rangle_\xi - \langle N_A \rangle_\xi^2, \\ \overline{F_A} = a_Q \log d_L(\ell) + b_Q + \delta_Q(\ell, L) \quad (24)$$

which probes the $U(1)$ current algebra rather than repeating the entropy fit. Figures 2(b) and 2(c) yield **[a_Q /CURRENT LEVEL, CORRELATION EXPONENT, AND WALL CONTOUR WEIGHT]**. The record-to-record variance of $\langle N_A \rangle_\xi$ is a different ensemble observable, familiar from charge-sharpening problems, and is not substituted for the within-trajectory quantum variance [63, 64]. The entanglement and charge contours then verify in real space that the logarithmic excess is carried by the programmed walls. Along the wall, we report density and current correlations through their decay exponents; a current correlator proportional to r^{-2} has scaling dimension $\Delta = 1$.

We next probe how this state is reached. Starting from the maximally mixed covariance $C_0 = 1/2$, we monitor the entropy or purity deficit $\Delta S_{\text{pur}}(t)$ for wall-on and wall-off circuits. Figures 2(d) and 2(e) show **[WALL POWER LAW, BULK RATE, AND TRANSVERSE LOCALIZATION]**. The tangent spectrum in Sec. VI relates this slow decay to **[MATCHED BOUNDARY TANGENT SECTOR]**.

IV. CHIRAL DYNAMICS OF THE DOMAIN WALL

Having established the critical content of the wall, we next determine its direction of propagation. A $c = 1$ logarithm alone does not fix $c_- = c_R - c_L$, so Fig. 3 brings together three complementary, orientation-sensitive probes.

A. Modular charge drift

For a trajectory covariance restricted to a wall region A , let $K_A = -\log \rho_A$ be the Gaussian modular Hamiltonian. A localized charge perturbation is evolved in modular time s ,

$$\delta n_\xi(y, s) = \text{Tr}[\delta \rho_{A,\xi} e^{\text{i}s K_{A,\xi}} n_y e^{-\text{i}s K_{A,\xi}}]. \quad (25)$$

We extract its centroid only while the packet is localized and before it meets the second wall or subsystem endpoint. We require the drift to reverse between oppositely oriented walls and to vanish in wall-off and exact-trivial controls. Figure 3(a) gives **[SIGNED MODULAR VELOCITIES AND WALL-OFF NULL RESULT]**. Because modular time is generated by the reduced state rather than the physical circuit, this drift measures the handedness of the state but is not a quantized transport coefficient.

FIGURE 2 PLACEHOLDER

- (a) Wall-excess entropy versus chord length, with exact- H dashed benchmark and fit-window audit.
- (b) Connected charge fluctuation and the independently extracted $U(1)$ coefficient; inset: wall correlator and decay exponent.
- (c) Entanglement/charge contour showing localization of the logarithmic contribution at the two programmed walls.
- (d) Wall-on versus wall-off purification over time and system size, including effective exponents.
- (e) Transverse profile of the residual late-time entropy or purity defect, identifying the wall as the slow sector.

FIG. 2. Critical state ensemble and slow wall purification. (a) Wall-excess entropy versus chord length, compared with the exact- H result on the same finite geometry. (b) Within-trajectory interval charge variance and its $U(1)$ current coefficient, with the wall correlation decay shown in the inset. (c) Entanglement and charge contours resolving the two programmed walls. (d) Purification from the maximally mixed state for wall-on and wall-off circuits. (e) Transverse profile of the late-time mixedness. Trajectory estimators are averaged with their Born-sampled weights, and dashed curves are benchmarks rather than fit constraints. Entanglement fixes $c_L + c_R$ for this geometry, not the sign of the chiral central charge.

B. Full Born response, including the score

The physical response of the Born ensemble includes both the change of the conditional state and the change in the probability of obtaining its record. For a localized potential impulse V and a trajectory observable $O_\xi[V]$,

$$\frac{\delta \langle O \rangle_V}{\delta V} \Big|_{V=0} = \left\langle \frac{\delta O_\xi[V]}{\delta V} \Big|_{V=0} \right\rangle_\xi + \langle \delta O_\xi s_\xi \rangle_\xi, \quad (26)$$

$$s_\xi = \frac{\delta \log p_\xi[V]}{\delta V} \Big|_{V=0}, \quad \delta O_\xi = O_\xi - \langle O \rangle. \quad (27)$$

The second term is the likelihood-score contribution. Omitting it gives the averaged conditional-state response, rather than the response of the physical Born ensemble. With $O = n(y, t)$, we Fourier transform along the wall and in circuit time. Figure 3(b) shows **[ONE-WAY RIDGE, v , ORIENTATION REVERSAL, AND SCORE CONTRIBUTION]**. A low-energy feature $\omega \simeq vk$ whose slope reverses with the wall orientation is the direct monitored-time propagation test in our analysis.

C. Fixed-record twist flow

Flux insertion supplies a complementary branchwise diagnostic. We freeze one realized circuit record ξ , insert a $U(1)$ twist ϕ through the periodic direction of that same realization, and follow the wall entanglement levels

$$\varepsilon_a(\phi) = \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)} \quad (28)$$

continuously for $0 \leq \phi \leq 2\pi$. Here $\nu_a(\phi)$ are the overlap-tracked eigenvalues of the wall-restricted trajectory covariance. The twist multiplies only controller modes crossing a fixed seam by $e^{i\phi}$; the stochastic couplings, update order,

and record are otherwise unchanged. Figure 3(c) yields **[ORIENTED SPECTRAL FLOW AND TRIVIAL CONTROL]**. Independently resampling the record at every ϕ would not define a continuous family and would have no Berry-holonomy interpretation. Fixed-record flow is therefore an auxiliary branchwise topology diagnostic, rather than the Born response of Eqs. (26) and (27); realizing it experimentally would require correlated record control.

V. UNIVERSAL DATA IN TRAJECTORY-RESOLVED RECORDS

We next ask whether the same wall theory can be identified from the classical measurement record alone. The relation

$$Z_\xi \equiv p_\xi \quad (29)$$

is only a starting point and does not by itself imply conformal structure. The directly sampled observable is the record self-information

$$I_\xi = -\log p_\xi, \quad H_{\text{rec}} = \langle I_\xi \rangle_\xi, \quad (30)$$

which is a Shannon entropy. To connect it to a statistical-mechanics free energy, we introduce the replica family

$$\mathcal{Z}_n(L, T) = \sum_\xi p_\xi^n, \quad -\partial_n \log \mathcal{Z}_n|_{n=1} = H_{\text{rec}}. \quad (31)$$

We derive the transfer representation and normalization of this replica derivative before assigning a conformal interpretation. The matched combination $H_{\text{rec}}^{\text{def}}$ uses wall-on, wall-off, and homogeneous circuits with identical local update counts. This subtraction cancels the common spacetime bulk contribution, while leaving a nonuniversal wall free-energy density that must be fitted together with the subextensive term.

FIGURE 3 PLACEHOLDER — THREE COMPLEMENTARY CHIRALITY PROBES

- (a) Modular charge-packet trajectories for the two wall orientations, with wall-off control.
 (b) Full Born response $\chi_{\text{Born}}(k, \omega)$, decomposed into conditional and likelihood-score terms; oriented low-energy ridge.
 (c) Fixed-record $0 \rightarrow 2\pi$ twist flow for paired wall orientations and matched trivial control.

FIG. 3. Three complementary tests of chirality. (a) Modular charge-packet motion in the two wall orientations and the wall-off control. (b) Momentum- and frequency-resolved Born susceptibility, separated into conditional-state and likelihood-score contributions. (c) Fixed-record twist flow for the paired wall orientations and a matched trivial circuit. Modular drift diagnoses directionality in the reduced state, the Born susceptibility measures physical monitored-time propagation, and the twist flow probes the topology of a continuous branch family.

Let $r = T/L$ be the temporal aspect ratio. Dividing the matched combination by the number of walls, the wall length, and the observation time defines

$$f_{\text{def}}(L, r) = \frac{H_{\text{rec}}^{\text{def}}(L, rL)}{N_{\text{wall}}L(rL)} = f_{\infty}(r) - \frac{A_{\text{rec}}(r)}{L^2} + \frac{b_4(r)}{L^4} + \dots \quad (32)$$

Equation (32) is the agnostic fit performed before any CFT conversion. Only after the replica normalization fixes a chiral periodic cylinder and the large- r limit is stable do we identify the per-wall amplitude as

$$A_{\text{rec}}(r) \xrightarrow{r \rightarrow \infty} \frac{\pi v_{\text{rec}} c_{\text{eff}}^{\text{rec}}}{12}. \quad (33)$$

The numerical coefficient is rederived if the implemented spacetime boundary conditions differ from this cylinder. We determine the nonuniversal velocity v_{rec} independently from transfer gaps or aspect-ratio collapse. Figures 4(a) and 4(b) give $[f_{\infty}, A_{\text{rec}}, \text{INDEPENDENT } v_{\text{rec}}, \text{AND } c_{\text{eff}}^{\text{rec}}]$. This effective central charge belongs to a replicated, generally nonunitary record theory and is not set equal to the trajectory-state value inferred from Fig. 2.

Charge-resolved record transfer eigenvalues provide a second finite-size observable. If $\Lambda_a(L)$ labels a sector of the record transfer problem, the expected form is

$$-\log \left| \frac{\Lambda_a(L)}{\Lambda_0(L)} \right| = \frac{2\pi v_{\text{rec}}}{L} x_a^{\text{rec}} + o(L^{-1}). \quad (34)$$

Figure 4(c) yields **[CHARGE SECTORS, COMMON VELOCITY, AND x_a^{rec}]**. Agreement of v_{rec} between Eqs. (33) and (34) prevents an arbitrary rescaling from manufacturing a central charge. The sector labels are defined from conserved charge or a controlled record deformation, rather than by sorting noisy eigenvalues after the fact.

This construction builds on trajectory thermodynamics and the trajectory-resolved statistical mechanics of monitored circuits, where record entropies and replicated transfer gaps can encode universal information [5, 6, 51, 55].

Here the transfer problem carries a defect tied to a programmable class-A interface. Related cross-entropy and postselection-free measurement strategies motivate operational estimators that avoid a histogram over exponentially many complete records [65, 66]. Figure 4(d) reports **[FINITE-SAMPLE ESTIMATOR BIAS, ERROR, AND SAMPLE SCALING]** against the exactly evaluated simulated likelihoods.

VI. TANGENT DYNAMICS AND SLOW PURIFICATION

Exact occupation projectors obstruct a global invertible transfer coordinate, but they do not obstruct linear response about a physical trajectory. In this section we construct the normalized tangent cocycle, extract its wall-localized slow sector, and connect that sector to purification from a maximally mixed state.

For a realized nonzero-probability branch, the normalized covariance update defines a map on its physical stratum,

$$G_{t+1} = \Phi_{\xi_t}[G_t]. \quad (35)$$

Linearizing this map propagates a perturbation without inverting the Kraus operator,

$$\delta G_{t+1} = D\Phi_{\xi_t}[G_t](\delta G_t) = \mathcal{J}_{t,\xi}^\dagger \delta G_t \mathcal{J}_{t,\xi}, \quad (36)$$

where the final factorization is taken on the support that survives the realized branch. Composing these differentials gives the chronological one-leg cocycle

$$\mathcal{T}_{T,\xi} = \mathcal{J}_{0,\xi} \mathcal{J}_{1,\xi} \cdots \mathcal{J}_{T-1,\xi}. \quad (37)$$

with the first-applied factor written leftmost. The product may lose rank, but its finite-time singular values and forward images remain well defined. We therefore use a stabilized forward filtration, which reduces to the familiar

FIGURE 4 PLACEHOLDER

- (a) Matched defect self-information density for several L and T/L , with f_∞ and the L^{-2} term shown separately.
 (b) Casimir/aspect-ratio collapse and extraction of $v_{\text{rec}} c_{\text{eff}}^{\text{rec}}$.
 (c) Charge-resolved record transfer gaps and common-velocity extraction of x_a^{rec} .
 (d) Record-only finite-sample estimator benchmark against exact simulated likelihoods.

FIG. 4. Universal data in trajectory-resolved records. (a) Matched defect self-information density and its L^{-2} coefficient for several temporal aspect ratios. (b) Cylinder collapse and the independent extraction of the record velocity and effective central charge. (c) Charge-resolved record transfer gaps and sector dimensions. (d) Finite-sample record-only estimator benchmarked against exact simulated likelihoods. The state central charge and the record effective central charge are displayed separately throughout.

Oseledec/Lyapunov construction when the factors are invertible [67, 68], without assuming backward subspaces through directions that have been projected out. The finite-time one-leg decay rates are

$$\gamma_a(T) = -\frac{1}{T} \log s_a(\mathcal{T}_{T,\xi}), \quad (38)$$

using a sign convention in which small positive γ_a denotes a slowly decaying one-leg direction. Because the covariance tangent is a congruence, its physical decay rates are the allowed pair sums $\gamma_a + \gamma_b$, rather than the individual one-leg rates.

The maximally mixed reference $G_0 = 0$ makes the connection to purification especially direct. For the full start-to- T cocycle accumulated along that same maximally mixed record, the exact endpoint relation

$$\mathcal{T}_{T,\xi}^\dagger \mathcal{T}_{T,\xi} = 4C_T(\mathbf{1} - C_T) \quad (39)$$

matches the tangent singular values to the remaining mixed occupations. This relation does not identify a separately sampled stationary pure-start QR spectrum. A finite one-leg lower edge $\gamma_* > 0$ produces exponential late-time purification. If the matched wall cocycle instead has a low-rate density $\rho(\gamma) \sim \gamma^\alpha$, the entropy deficit scales as

$$\Delta S_{\text{pur}}(t) \sim \int_0^\infty d\gamma \rho(\gamma) e^{-2\gamma t} \sim t^{-(\alpha+1)}. \quad (40)$$

Figure 5(d) gives [**α AND PARAMETER-FREE PURIFICATION EXPONENT**]. The exponent extracted from the tangent density is compared with Fig. 2(d) without independently refitting the purification curve to force agreement.

We now characterize the slow tangent sector itself. After taking the long-time limit at each fixed transverse size, Fig. 5(b) gives [**WALL-ON GAP SCALING AND FINITE WALL-OFF GAP**]. Because several slow directions can become nearly degenerate, Fig. 5(c) displays

the projector density of the complete low-exponent subspace rather than a single QR vector, yielding [**WALL WEIGHT AND LOCALIZATION LENGTH**]. The trajectory-conditioned spectrum is finally compared with the unconditional covariance map in Fig. 5(e), where we find [**CONDITIONAL-UNCONDITIONAL GAP COMPARISON**]. These tests place the present rank-losing circuit alongside monitored Lyapunov constructions while keeping the branch tangent distinct from an invertible logarithmic generator [18, 19].

For visualizing the finite-time singular directions, we also use the Hermitian log-stretch generator

$$P_{T,\xi}^\Lambda = \frac{1}{2T} \log(\mathcal{T}_{T,\xi} \mathcal{T}_{T,\xi}^\dagger) \quad (41)$$

on the surviving support. The object $P_{T,\xi}^\Lambda$ is not a density matrix and need not be positive. Its eigenvectors are the singular directions of one finite-time trajectory product, not eigenvectors obtained by averaging trajectory vectors. Any quenched average is therefore applied either to this logarithmic object or to continuously tracked exponents, with the order of operations stated explicitly.

The same singular structure admits a scattering-network interpretation. A regulated Gaussian Choi relation separates into an invertible quotient core and occupation caps that pin exactly filled or empty directions, while fresh gain/loss ancillas act as physical reservoir ports. Chronological composition eliminates internal ports by a Schur complement, and the caps retain the rank information discarded by an ordinary transfer inverse. Removing the regulator leaves the normalized tangent core in Eq. (36); Appendix C4 gives the cap bookkeeping, particle/hole symmetry, and regular-transfer limit.

FIGURE 5 PLACEHOLDER

- (a) Stabilized tangent-product workflow for exact projectors and gain/loss branches.
- (b) Wall-on and wall-off minimum tangent gaps versus transverse size, including zero-intercept and alternative finite-size fits.
- (c) Globally normalized projector density of the complete near-marginal subspace, averaged over Born trajectories.
- (d) Low-exponent density and parameter-free comparison with the purification curve of Fig. 2.
- (e) Trajectory-conditioned result versus the mixed unconditional covariance/channel control.

FIG. 5. Tangent mechanism for slow wall purification. (a) Stabilized product of normalized branch differentials. (b) Long-time-extrapolated wall-on and wall-off tangent gaps versus transverse size. (c) Basis-independent projector density of the complete near-degenerate slow subspace, averaged over Born trajectories. (d) Low-exponent density and its parameter-free prediction for purification. (e) Comparison with the unconditional covariance map. Component-resolved plots preserve their relative physical weights.

VII. ROBUSTNESS AND ENSEMBLE CONTROLS

We finally test whether the wall signatures persist away from the ideal perfect-correction limit. We introduce **[IMPLEMENTED IMPERFECTION]** and follow the bulk marker, wall entropy coefficient, tangent gap, and one orientation-sensitive response [Appendix Fig. 11]. This analysis gives **[ROBUSTNESS INTERVAL OR THRESHOLD SCALING]**. Feedback-driven entanglement, steering, control, and absorbing-state transitions are already known in interacting and random adaptive circuits [32, 33, 69–71]; transport-limited frustration-free control provides a complementary preparation problem [35]. Our purpose here is the stability of the same topological defect; a complete absorbing-transition analysis would additionally require an order parameter, finite-size crossings, and a dynamical exponent.

The earlier Bhuiyan–Pan–Jian noise study suggests a natural endpoint for this robustness window [20]. Once noise destroys the bulk dynamical topological distinction or closes the relevant purity gap, a critical wall is no longer topologically required, although accidental boundary criticality is not excluded. The measured relation between the two scales is **[BULK–WALL THRESHOLD RELATION]**. A boundary crossover or preemptive wall transition can therefore occur before the bulk threshold; coincidence is not imposed in the analysis.

A matched postselected control addresses a different question. For the same local schedule, Appendix H compares Born sampling with a forced favorable record and, where meaningful, an explicitly designed imaginary-time filter. The initial overlap, Trotter step, normalization, and success probability are retained so that the comparison isolates which wall and record observables require the physical Born ensemble.

The averaged channel provides the final ensemble control. Averaging Gaussian Kraus trajectories need not produce a Gaussian many-body channel, so we

use closed covariance formulas only where closure is explicitly demonstrated. On that well-defined sector, Fig. 5(e) gives **[AVERAGED-CHANNEL GAP OR OCCUPATION-BRANCH RESULT]**. A half-occupied mode or linear occupation branch is reported as a property of the mixed channel and is not identified with the trajectory CFT.

VIII. CONCLUSION AND OUTLOOK

A. Summary of results

In this work, we have used the perfect-correction limit of an adaptive Chern circuit to investigate the universal dynamics of a programmable domain wall. Born sampling does not prepare one corrected wave function: it generates a charge-fluctuating ensemble of inequivalent Gaussian states. Across that ensemble, the wall exhibits **[FINAL STATE-CFT RESULT]**, while the modular, response, and twist probes establish **[FINAL CHIRALITY RESULT]**. The two microscopic wall constructions and the exact-Hamiltonian benchmarks separate these boundary properties from details of the finite-support controller.

The singular tangent formalism supplies the dynamical mechanism. The differential of the normalized covariance map remains meaningful through rank-changing occupation projectors and gives **[FINAL TANGENT AND PURIFICATION RESULT]**. The replicated Born weights independently yield **[FINAL RECORD RESULT]**, with the state and record central charges kept as distinct quantities. Comparison with the unconditional covariance map gives **[FINAL CHANNEL COMPARISON]**. Together, these observations define a three-layer bulk–boundary structure: a monitored class-A bulk, a chiral critical wall in its physical Born trajectories, and a defect theory in the probability distribution of those trajectories.

This structure distinguishes the present problem from

adaptive preparation of a common target state. Local measurement and feedforward can compile conformal and chiral states, including through parton resources and multiscale adaptive circuits [28]. Here those ingredients instead act repeatedly and locally, leaving the realized record and total charge stochastic. The earlier Bhuiyan–Pan–Jian construction supplies the postselection-free bulk controller, programmable wall, and slow-purification observation [20]; the additional result is the identification of the wall theory across state, orientation-sensitive, tangent, record, and channel observables.

The tangent and record constructions likewise extend, rather than replace, existing monitored-fermion descriptions. Gaussian tensor-network and transfer approaches relate monitored dynamics to localization, and Lyapunov classifications identify topological boundary sectors [15, 17–19]. Our normalized cocycle carries this analysis through exact rank loss and connects its low-exponent density quantitatively to purification. Non-Hermitian or no-click Hamiltonians remain useful descriptions of selected branches, dissipative preparation constrains the unconditional channel, and continuous counting theory characterizes physical stochastic records [49, 52, 53, 72]. The present contribution is their conjunction at a programmable two-dimensional Chern interface, together with a replica-calibrated defect construction rather than the bare relation $Z_\xi = p_\xi$. Monitored free-fermion criticality supplies both the nearby phenomena and the necessary large-scale crossover cautions [22, 23, 27, 73]. Interactive feedback and control provide a distinct setting in which entanglement and absorbing transitions can coexist [33, 69, 71].

B. Outlook and future directions

The first extension is to interacting adaptive circuits. Covariance closure is then lost, but the distinction among a selected branch, the Born ensemble, the record distribution, and the unconditional channel remains. Determining which parts of the tangent and record constructions survive beyond Gaussian dynamics would provide a route toward interacting chiral defect theories.

A second direction is the competition between adaptive correction and noise or an absorbing process. The robustness analysis here follows one imperfection across the bulk marker and wall observables. A complete transition theory would determine whether the wall loses criticality at the bulk topological threshold, crosses over earlier, or meets a separate absorbing or control transition, with independent order parameters and dynamical scaling.

Finally, the fresh-ancilla realization gives a direct quantum-simulation target. Refreshed occupied and empty modes, midcircuit occupation measurements, local feedforward, and fermionic exchange operations implement the ideal update using the BPJ route. The experiment would prepare and sample a universal charge-fluctuating ensemble with a programmable defect, rather

than certify one Chern wave function. Its most distinctive observable may be the comparison between universal quantities extracted from the classical record and those requiring trajectory-state tomography. Establishing the relative sample and hardware cost of those two routes is an important operational benchmark for future native-fermion processors [40, 74–76].

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Appendix A: Microscopic instrument and exact perfect correction

This appendix fixes the microscopic convention used throughout the paper. In the main text, a controller channel carries the BPJ label $j = (\mathbf{r}, \nu, \sigma)$, with $\nu = A, B$, $\sigma = \pm$, localized operator $\hat{\chi}_{\mathbf{r}, \nu, \sigma}$, and occupation $\hat{\mathcal{N}}_{\mathbf{r}, \nu, \sigma} = \hat{\chi}_{\mathbf{r}, \nu, \sigma}^\dagger \hat{\chi}_{\mathbf{r}, \nu, \sigma}$. The coefficient column of that operator is denoted by the corresponding plain symbol χ_j below. Thus the generic notation a_χ and n_χ used in the derivations is precisely $\hat{\chi}_j$ and $\hat{\mathcal{N}}_j$ for a selected BPJ channel, not an additional fermion mode. We consider N complex fermion modes with

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = 0, \quad (\text{A1})$$

and use the number-conserving two-point matrix and centered covariance

$$C_{ij} = \langle c_i^\dagger c_j \rangle, \quad G = 2C - \mathbf{1}. \quad (\text{A2})$$

Thus $0 \leq C \leq \mathbf{1}$, while the eigenvalues of G lie in $[-1, 1]$. A pure Slater determinant obeys $C^2 = C$, equivalently $G^2 = \mathbf{1}$. All formulas below act on the physical one-particle space; an orbital projector is denoted by $P_\chi = \chi \chi^\dagger$, with $\chi^\dagger \chi = 1$.

1. Occupation measurement and branch probabilities

For the normalized orbital

$$a_\chi = \sum_j \chi_j c_j, \quad n_\chi = a_\chi^\dagger a_\chi, \quad (\text{A3})$$

the two projectors are

$$\Pi_1 = n_\chi, \quad \Pi_0 = \mathbf{1} - n_\chi = a_\chi a_\chi^\dagger. \quad (\text{A4})$$

It is convenient to write $\eta_m = 2m - 1$, so that $\eta_m = +1$ for an occupied result and $\eta_m = -1$ for an empty result. On a Gaussian state the Born probabilities are

$$p_m(G) = \text{Tr}(\rho \Pi_m) = \frac{1 + \eta_m g_\chi}{2}, \quad g_\chi = \chi^\dagger G \chi. \quad (\text{A5})$$

Only outcomes with $p_m > 0$ define normalized branches. A formal pole at $p_m = 0$ is therefore a zero-probability branch, not a failure of the Gaussian update.

Let $s \in \{0, 1\}$ be the target occupation assigned by the controller. For the upper-band controller $s = 0$, while for the lower-band controller $s = 1$. Perfect correction is represented by the two system Kraus operators

$$\begin{array}{c|cc} & m=0 & m=1 \\ \hline s=0 & K_{0|0} = \Pi_0 & K_{1|0} = a_\chi \\ s=1 & K_{0|1} = a_\chi^\dagger & K_{1|1} = \Pi_1 \end{array} \quad (\text{A6})$$

These operators satisfy, branch by branch,

$$\begin{aligned} K_{m|s}^\dagger K_{m|s} &= \Pi_m, & \Pi_s K_{m|s} &= K_{m|s}, \\ \sum_m K_{m|s}^\dagger K_{m|s} &= \mathbf{1}. \end{aligned} \quad (\text{A7})$$

The first identity shows that correction does not alter the Born probability. The second shows that the measured orbital has the target occupation after every nonzero branch. Randomness remains in the conditional state of the overlapping orbitals; perfect correction does not collapse the full circuit to a deterministic wavefunction.

2. Fresh-ancilla dilation and deterministic fSWAP correction

Introduce one fresh ancillary mode d , and define

$$\begin{aligned} U_{\text{fSWAP}} &= \text{fSWAP}(a_\chi, d) = \exp\left[\mathbf{i}\frac{\pi}{2}(a_\chi^\dagger - d^\dagger)(a_\chi - d)\right] \\ &= \mathbf{1} - n_\chi - n_d + a_\chi^\dagger d + d^\dagger a_\chi. \end{aligned} \quad (\text{A8})$$

It leaves $|00\rangle$ invariant, exchanges $|01\rangle$ and $|10\rangle$, and gives the fermionic exchange sign on $|11\rangle$. After the occupation measurement, the physical mode is already in the definite state $|m\rangle_\chi$. If $m \neq s$, the ancilla is prepared in $|s\rangle_d$ and fSWAP is applied. Consequently,

$$|m\rangle_\chi |s\rangle_d \xrightarrow{U_{\text{fSWAP}}} |s\rangle_\chi |m\rangle_d, \quad m \neq s. \quad (\text{A9})$$

The outgoing ancilla has a known occupation and factorizes from the physical system. It may therefore be discarded, or locally reset and reused, without postselection. Taking the appropriate ancilla matrix element reproduces the odd entries in Eq. (A6),

$$\langle 0|U_{\text{fSWAP}}|1\rangle_d = a_\chi^\dagger, \quad \langle 1|U_{\text{fSWAP}}|0\rangle_d = a_\chi, \quad (\text{A10})$$

but these matrix elements are a derivation of the reduced Kraus map, not an instruction to postselect the ancilla. Total charge is conserved on the enlarged system-ancilla Hilbert space.

3. Mixed fresh ancilla and Markovian bath control

A distinct control replaces the target Fock-state ancilla by a freshly drawn mixed mode

$$\rho_a(\bar{n}_a) = (1 - \bar{n}_a)|0\rangle\langle 0|_a + \bar{n}_a|1\rangle\langle 1|_a, \quad 0 \leq \bar{n}_a \leq 1. \quad (\text{A11})$$

Apply the same fSWAP as in Eq. (A8), now between the system orbital selected by $P = P_\chi$ and this ancilla, and trace out the outgoing ancilla. With $Q = \mathbf{1} - P$, the reduced system covariance is

$$\boxed{\mathcal{B}_{\bar{n}_a}(G) = QGQ + (2\bar{n}_a - 1)P.} \quad (\text{A12})$$

Thus the selected system mode is replaced by the bath occupation and all of its incoming correlations with the complementary modes leave with the discarded ancilla.

Equivalently, one may first sample the classical preparation label $a \in \{0, 1\}$ of Eq. (A11). Writing $\eta_a = 2a - 1$, the two conditional replacement maps are

$$\begin{aligned} G'_+ &= QGQ + P, & p_+ &= \bar{n}_a, \\ G'_- &= QGQ - P, & p_- &= 1 - \bar{n}_a, \end{aligned} \quad (\text{A13})$$

whose classical average is Eq. (A12). These branches are generally mixed even for a pure incoming system, because tracing the outgoing ancilla discards any initial entanglement of the selected mode. They remain pure only when the input is pure and the selected mode is already uncorrelated with its complement, $QGP = 0$. Alternatively, conditioning on an occupation measurement of the outgoing ancilla restores a pure-state measurement trajectory but gives the nonlinear conditional map of Eq. (A14), not Eq. (A13).

Equation (A12) is therefore a Markovian bath or noise control, not the production perfect-correction rule: the latter chooses the definite target occupation s after observing the system outcome. Fresh preparation (or a complete reset) is essential. Reusing the outgoing ancilla without resetting it feeds the previous system mode back into later steps and produces a non-Markovian instrument.

4. Conditional covariance map

Let $P = P_\chi$, $Q = \mathbf{1} - P$, and retain only a nonzero outcome m . The conditional measurement followed by perfect reset to s acts on the centered covariance as

$$\boxed{\Phi_{m|s}(G) = Q(\mathbf{1} + \eta_m GP)^{-1} GQ + \eta_s P.} \quad (\text{A14})$$

The inverse is only a rank-one resolvent. The Sherman-Morrison form makes the probability denominator explicit,

$$(\mathbf{1} + \eta_m GP)^{-1} = \mathbf{1} - \frac{\eta_m GP}{1 + \eta_m g_\chi}. \quad (\text{A15})$$

Perfect-correction instrument and its local dilation

Panel (a): occupation measurement of a finite-support orbital, with the two Born outcomes. Panel (b): the four entries of Eq. (A6). Panel (c): deterministic gain and loss realized by an occupied or empty fresh ancilla followed by fSWAP. The final artwork should distinguish the retained physical record from a postselected ancilla port and should indicate that the outgoing ancilla is factorized and recyclable.

FIG. 6. Perfect-correction instrument and its local dilation. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

Equation (A14) applies equally to pure and mixed Gaussian inputs. The last term is set by the target, whereas the state-dependent contraction of the orthogonal complement is selected by the actual measurement result.

For reference, the unconditional perfect-correction channel for this one orbital is especially simple. Summing the unnormalized branches gives the replacement channel

$$\mathcal{E}_s(\rho) = \sum_m K_{m|s} \rho K_{m|s}^\dagger = |s\rangle\langle s|_\chi \hat{\otimes} \text{Tr}_\chi \rho, \quad (\text{A16})$$

where $\hat{\otimes}$ denotes the graded fermionic product. Its affine covariance action is

$$\mathbb{E}_m[\Phi_{m|s}(G)] = QGQ + \eta_s P. \quad (\text{A17})$$

This form is used on the parity-even, number-conserving states of the present protocol; the graded tensor factorization is part of the statement. The nonlinear conditioned map and the linear averaged channel must not be interchanged in trajectory observables.

For an ordered record $\xi = (m_1, \dots, m_D)$, with prescribed targets s_t and orbitals χ_t , define

$$G_t = \Phi_{m_t|s_t}(G_{t-1}). \quad (\text{A18})$$

The complete Born weight is the chronological product

$$p_\xi(G_0) = \prod_{t=1}^D \frac{1 + \eta_{m_t} \chi_t^\dagger G_{t-1} \chi_t}{2}. \quad (\text{A19})$$

All later factors are evaluated on the state produced by the earlier outcomes. Evaluating every probability at G_0 , or attaching additional weights to already Born-sampled trajectories, would give a different ensemble.

The chronological channel convention is also part of the model. At each active unit cell, the manuscript follows BPJ and visits

$$\begin{aligned} \mathcal{Q}_{\text{BPJ}} &= ((A, -), (A, +), (B, -), (B, +)), \\ (s_q)_{q \in \mathcal{Q}_{\text{BPJ}}} &= (1, 0, 1, 0). \end{aligned} \quad (\text{A20})$$

Equivalently, the circuit fills the lower OW mode and then depletes the upper OW mode for $\nu = A$, before repeating the same pair for $\nu = B$. Because overlapping occupations do not generally commute, a word with the same four labels in another order defines a different finite-range instrument and cannot be identified with Eq. (A20) by relabeling its output.

Appendix B: Finite-support controllers and programmed domain walls

This appendix supplies the Fourier conventions, the effect of real-space windowing, and the implementation details suppressed in Sec. II. The distinction among the target-band Chern number, the winding of a windowed mode's overlap with that band, and the Chern number of the compact mode itself is particularly important.

1. Uniform parent and projected frame

On a finite periodic lattice, the real-space band projector associated with Eq. (1) is

$$P_b(\mathbf{r} - \mathbf{r}'; \alpha) = \frac{1}{N_x N_y} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} P_b(\mathbf{k}; \alpha). \quad (\text{B1})$$

For the orthonormal cell-local trials $\tau_A = (1, 1)^\top / \sqrt{2}$ and $\tau_B = (1, -1)^\top / \sqrt{2}$, define the conventional projected one-particle OW vector. The appendix uses a generic band label $b = \pm$; it is the sign $\sigma = \pm$ in the BPJ channel label of the main text:

$$|\widetilde{W}_{\mathbf{R}, \nu, b}\rangle = P_b(\alpha) |\mathbf{R}, \tau_\nu\rangle. \quad (\text{B2})$$

The corresponding unnormalized annihilation operator is

$$\begin{aligned} \hat{\chi}_{\mathbf{R}, \nu, b} &= \sum_{\mathbf{r}, \mu} \langle \widetilde{W}_{\mathbf{R}, \nu, b} | \mathbf{r}, \mu \rangle c_{\mathbf{r}\mu} \\ &= \frac{1}{\sqrt{N_x N_y}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}} \tau_\nu^\dagger P_b(\mathbf{k}; \alpha) \mathbf{c}_{\mathbf{k}}. \end{aligned} \quad (\text{B3})$$

The normalized BPJ operator is

$$\hat{\chi}_{\mathbf{R},\nu,b}^{(\infty)} = \frac{\tilde{\chi}_{\mathbf{R},\nu,b}}{\|\tilde{W}_{\mathbf{R},\nu,b}\|}. \quad (\text{B4})$$

Writing $N_c = N_x N_y$, the vector stored by the numerical covariance code is the coefficient column

$$\tilde{\chi}_{\mathbf{R}\nu b}(\mathbf{r}, \mu) = \frac{1}{N_c} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} [P_b(\mathbf{k}; \alpha) \tau_\nu]_\mu^*. \quad (\text{B5})$$

It equals $\langle \tilde{W}_{\mathbf{R},\nu,b} | \mathbf{r}, \mu \rangle$. This conjugation is why the filled-lower-band covariance in the convention $C_{ij} = \langle c_i^\dagger c_j \rangle$ is $C_{\text{CI}}(\mathbf{k}) = P_-^*(\mathbf{k})$. The physical one-particle projector is recovered as $C_{\text{CI}}^\top = P_-$; hence the target Chern number remains +1 when the same convention is used in the marker. The transpose must be retained when comparing analytic one-particle kets with code covariance projectors.

The word “overcomplete” is made precise by the unnormalized tight-frame identity

$$\begin{aligned} \sum_{\mathbf{R},\nu} |\tilde{W}_{\mathbf{R},\nu,b}\rangle \langle \tilde{W}_{\mathbf{R},\nu,b}| &= P_b \left(\sum_{\mathbf{R},\nu} |\mathbf{R}, \tau_\nu\rangle \langle \mathbf{R}, \tau_\nu| \right) P_b \\ &= P_b. \end{aligned} \quad (\text{B6})$$

There are two projected vectors per cell for a one-band subspace of one state per cell. They span that subspace without constituting an orthogonal Wannier basis. After normalizing every nonzero full-support mode, the filled lower band obeys the common target relations in Eq. (9).

2. Atomic large-mass limit

The large positive trivial mass has a simple gauge-independent limit. For the Bloch vector in Eq. (1),

$$\mathbf{d}(\mathbf{k}; \alpha) = (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \quad (\text{B7})$$

the spectral projectors may be written

$$P_\pm(\mathbf{k}; \alpha) = \frac{1}{2} \left[\mathbf{1} \pm \hat{\mathbf{d}}(\mathbf{k}; \alpha) \cdot \boldsymbol{\sigma} \right]. \quad (\text{B8})$$

At fixed momentum,

$$\hat{\mathbf{d}}(\mathbf{k}; \alpha) = \hat{\mathbf{z}} + O(\alpha^{-1}), \quad P_\pm(\mathbf{k}; \alpha) = \frac{1}{2} (\mathbf{1} \pm \sigma_z) + O(\alpha^{-1}) \quad (\text{B9})$$

as $\alpha \rightarrow +\infty$. The leading correction is momentum dependent and off diagonal in the orbital basis, while the diagonal correction begins at higher order.

Fourier transforming Eq. (B9) gives

$$P_\pm(\mathbf{r} - \mathbf{r}'; \alpha) = \delta_{\mathbf{r},\mathbf{r}'} \frac{1 \pm \sigma_z}{2} + O(\alpha^{-1}). \quad (\text{B10})$$

Consequently the normalized projected trial vectors of Eq. (B2) collapse onto the corresponding onsite orbital

whenever the trial has nonzero overlap with that orbital. The choice $\alpha_{\text{triv}} = 30$ is therefore a controlled proxy for an atomic trivial exterior. The support-terminated controller of Eq. (B22) is nevertheless a separate active boundary condition: masking a topological OW frame and omitting exterior centers is not identical to evolution with any finite value of α_{triv} , even though the infinite-mass limit motivates the hard-boundary interpretation.

3. Truncate-then-renormalize rule

For integer shell radius n , let

$$M_{\mathbf{R}}^{(n)}(\mathbf{r}) = \mathbf{1} [|d_{N_x}(x, R_x)| \leq n \text{ and } |d_{N_y}(y, R_y)| \leq n]. \quad (\text{B11})$$

The measured operator coefficient is obtained by applying every geometric mask and only then normalizing,

$$\chi_{\mathbf{R}\nu b}^{(n)} = \frac{M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}}{\|M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}\|}, \quad a_{\mathbf{R}\nu b}^{(n)} = \sum_{\mathbf{r}, \mu} \chi_{\mathbf{R}\nu b}^{(n)}(\mathbf{r}, \mu) c_{\mathbf{r}\mu}. \quad (\text{B12})$$

Thus $n = 1$ is a 3×3 -cell square with at most 18 microscopic modes. The special half-shell option is instead the five-cell cross $\{(0, 0), (\pm 1, 0), (0, \pm 1)\}$. Normalization is performed independently for every $(\mathbf{R}, \nu, b)$. Windowing introduces out-of-band weight, so the compact occupations cease to be a mutually compatible set of exact stabilizers even though the controller remains strictly local.

4. Windowed overlap and the surviving obstruction

Let $|u_b(\mathbf{k})\rangle$ be a normalized Bloch eigenstate and $f_{\nu b}(\mathbf{k}) = \langle u_b(\mathbf{k}) | \tau_\nu \rangle$ its trial-state form factor. Multiplication by a real-space window becomes a twisted convolution. For a mode centered at the origin, its overlap with the target band is

$$f_{\nu b}^{(n)}(\mathbf{k}') = \frac{1}{N_x N_y} \sum_{\mathbf{k}} \tilde{M}_n(\mathbf{k}' - \mathbf{k}) f_{\nu b}(\mathbf{k}) \langle u_b(\mathbf{k}') | u_b(\mathbf{k}) \rangle, \quad (\text{B13})$$

where the square window has

$$\begin{aligned} \tilde{M}_n(\mathbf{q}) &= D_n(q_x) D_n(q_y), \\ D_n(q) &= \sum_{j=-n}^n e^{-iqj} = \frac{\sin[(n + \frac{1}{2})q]}{\sin(q/2)}. \end{aligned} \quad (\text{B14})$$

Restoring a nonzero center adds only the translation phases displayed in the full derivation of the windowed controller. Dividing the compact mode by its real-space norm multiplies Eq. (B13) by a positive constant and therefore does not change its zeros or phase winding.

For a small positively oriented loop Γ enclosing an isolated zero, define

$$\nu_{\text{ov}}^{(n)}(\Gamma) = \frac{1}{2\pi i} \oint_{\Gamma} d\mathbf{k} \cdot \nabla_{\mathbf{k}} \log f_{\nu,-}^{(n)}(\mathbf{k}). \quad (\text{B15})$$

For the unwindowed form factor, the convention of Eq. (1) gives $\mathcal{C}[P_-] = -\nu_{\text{ov}}^{(\infty)}$. At $\alpha = 1$, the A trial has its unwindowed zero at $\mathbf{k}_0/\pi = (0.5, 0)$. Direct evaluation of the square-window convolution gives

n_{shell}	support	$\ M^{(n)}\widetilde{W}\ ^2/\ \widetilde{W}\ ^2$	$\mathbf{k}_*^{(n)}/\pi$
1	3×3	0.992155	(0.4922, 0)
2	5×5	0.999137	(0.4925, 0)

(B16)

For both rows, the descendant zero has $\nu_{\text{ov}}^{(n)} = -1$. The zero is displaced rather than filled. Its local form remains $f^{(n)}(\mathbf{k}_* + \mathbf{p}) = A_n(p_x - ip_y) + O(p^2)$, which fixes the reported winding. Equation (B16), rather than the retained norm alone, is the finite-window check relevant to the controller.

For a reproducible discrete audit, sample a loop $\Gamma = (\mathbf{k}_0, \dots, \mathbf{k}_{N_\Gamma} = \mathbf{k}_0)$ and compute

$$\nu_{\text{loop}}^{(n)} = \frac{1}{2\pi} \sum_{j=0}^{N_\Gamma-1} \text{Arg} \left[\frac{f^{(n)}(\mathbf{k}_{j+1})}{f^{(n)}(\mathbf{k}_j)} \right]. \quad (\text{B17})$$

The loop must avoid the zero and remain within a region containing no other zeros. The result is checked under increasing momentum resolution, several loop radii, and smooth gauge changes of $u_-(\mathbf{k})$; the phase ratios make the final integer gauge invariant when the loop is closed consistently.

The overlap winding is not the Chern number of the compact-support spinor. Writing its Bloch transform as the two-component trigonometric polynomial

$$|\psi_n(\mathbf{k})\rangle = \sum_{\mathbf{r} \in \text{supp } M^{(n)}} e^{-i\mathbf{k} \cdot \mathbf{r}} C_{\mathbf{r}} |\tau_n\rangle, \quad C_{\mathbf{r}} = \int_{\text{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\mathbf{q} \cdot \mathbf{r}} P_-(\mathbf{q}), \quad (\text{B18})$$

one sees that, whenever $\psi_n(\mathbf{k}) \neq 0$ throughout the Brillouin torus, its normalized version is a smooth periodic global section. Direct minimization for the $n = 1$ mode gives $\min_{\mathbf{k}} \|\psi_1(\mathbf{k})\| = 5.106 \times 10^{-2}$, so this condition is satisfied. Hence

$$\mathcal{C}_{\text{target}} = +1, \quad \nu_{\text{ov}}^{(1)} = -1, \quad \mathcal{C}_{\text{compact}}^{(1)} = 0. \quad (\text{B19})$$

A coarse Fukui–Hatsugai–Suzuki discretization of an almost-vanishing compact spinor can cross its admissibility bound and return a spurious integer. We therefore apply FHS to the reconstructed ket-space target projector $C_{\text{CI}}^\top = P_-$, and evaluate real-space markers with the same transpose/orientation convention. The loop winding instead diagnoses the overlap of that target with the compact controller; the two estimators are not identified [58].

5. Exact implementation of the two walls

Let $\mathcal{R}_{\text{top}} = \{\mathbf{R} : x_L \leq R_x \leq x_R\}$, with the endpoints stated explicitly for every geometry. The explicit-interface

circuit uses

$$\alpha_{\mathbf{R}} = \begin{cases} \alpha_{\text{top}}, & \mathbf{R} \in \mathcal{R}_{\text{top}}, \\ \alpha_{\text{triv}}, & \mathbf{R} \notin \mathcal{R}_{\text{top}}, \end{cases} \quad \chi_{\mathbf{R}\nu b}^{(n), \text{exp}} = \frac{M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}(\alpha_{\mathbf{R}})}{\|M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}(\alpha_{\mathbf{R}})\|}. \quad (\text{B20})$$

All centers remain active. Each center uses a projector from one of two *uniform* parents, and its compact tail may cross the interface. It is therefore inaccurate to describe Eq. (B20) as the spectral frame of a single spatially inhomogeneous Hamiltonian.

The code constructs the more general same-region mask

$$D_{\mathbf{R}}(\mathbf{r}) = \Theta_{\text{top}}(\mathbf{R})\Theta_{\text{top}}(\mathbf{r}) + [1 - \Theta_{\text{top}}(\mathbf{R})][1 - \Theta_{\text{top}}(\mathbf{r})]. \quad (\text{B21})$$

The support-terminated protocol studied in the paper then restricts the active set to $\mathbf{R} \in \mathcal{R}_{\text{top}}$, for which $D_{\mathbf{R}}(\mathbf{r}) = \Theta_{\text{top}}(\mathbf{r})$, and uses

$$\chi_{\mathbf{R}\nu b}^{(n), \text{term}} = \frac{D_{\mathbf{R}} M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}(\alpha_{\text{top}})}{\|D_{\mathbf{R}} M_{\mathbf{R}}^{(n)} \tilde{\chi}_{\mathbf{R}\nu b}(\alpha_{\text{top}})\|}, \quad \mathbf{R} \in \mathcal{R}_{\text{top}}. \quad (\text{B22})$$

Both $D_{\mathbf{R}}$ and $M_{\mathbf{R}}^{(n)}$ are diagonal real-space multiplication operators, so they commute. The order in Eq. (B22) is merely notational: the product mask defines the orbital before the circuit begins, and the result is normalized once. Exterior canonical orbitals are measured once to prepare a Born-sampled onsite product state and are omitted from subsequent OW cycles. The exterior outcome word ζ , its probability p_ζ , and the conditional state $\rho_{0|\zeta}$ are stored separately; the complete trajectory weight is $p_{\zeta, \xi|\pi} = p_\zeta p_{\xi|\pi, \zeta}$. A two-sided hard-truncation protocol that continues to update exterior centers is a different circuit and is not pooled with the support-terminated data.

On a periodic lattice the inclusive cell interval $[x_L, x_R]$ places the physical interfaces on the bonds $x_L - \frac{1}{2}$ and $x_R + \frac{1}{2}$. Quoted wall widths must use these explicit endpoints; implementation-dependent default fractions of N_x are not part of the model definition. Finally, the exact- H benchmark is obtained by diagonalizing a genuinely inhomogeneous single-particle Hamiltonian. It tests the expected finite-geometry boundary theory but is neither Eq. (B20) nor Eq. (B22).

Appendix C: Singular normalized tangent cocycle

The appropriate linear response of a conditioned trajectory is the derivative of the normalized covariance map. Differentiating Eq. (A14) at a nonzero branch gives

$$D\Phi_{m|s}|_G[H] = J_m(G)^\dagger H J_m(G), \quad (\text{C1})$$

with the half-Jacobian

$J_m(G) = Q - \eta_m \frac{PGQ}{1 + \eta_m g_\chi}.$

(C2)

Finite window and overlap obstruction

Panel (a): real-space weight of the unwindowed $A-$ mode and the 3×3 window. Panel (b): $|f_{A,-}^{(n)}(k_x, 0)|$ for the unwindowed, $n = 1$, and $n = 2$ modes, marking the displaced zeros. Panel (c): phase circulation on loops surrounding the $n = 1$ zero. Panel (d): target-band FHS invariant, overlap winding, and compact-spinor FHS result shown as three explicitly different diagnostics.

FIG. 7. Finite window and overlap obstruction. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

The target s does not enter this derivative: correction replaces the measured mode by a constant covariance, while the realized outcome controls the response of its complement. In particular,

$$J_m(G)\chi = 0. \quad (\text{C3})$$

Exact projectors therefore produce a semi-invertible co-cycle. No inverse of the accumulated product is required.

For a normalized many-body map $\rho' = K\rho K^\dagger/p_K$, the same statement follows from the quotient rule

$$\delta\rho' = \frac{1}{p_K} [K\delta\rho K^\dagger - \rho' \text{Tr}(K\delta\rho K^\dagger)]. \quad (\text{C4})$$

The subtraction removes the norm-changing direction. It is precisely this Born normalization that distinguishes $J_m(G)$ from a bare one-particle transfer matrix.

1. Chronological order and QR accumulation

Along a fixed record, let $J_t = J_{m_t}(G_{t-1})$. Repeated use of Eq. (C1) yields

$$\delta G_D = \tilde{J}_D^\dagger \delta G_0 \tilde{J}_D, \quad \boxed{\tilde{J}_D = J_1 J_2 \cdots J_D}. \quad (\text{C5})$$

The first-applied factor is leftmost in $\tilde{J}_D$. A column QR implementation propagates the adjoint factors in the reverse-looking, but equivalent, order:

$$Y_t = J_t^\dagger Q_{t-1} = Q_t R_t, \quad \tilde{J}_t^\dagger Q_0 = Q_t R_t R_{t-1} \cdots R_1. \quad (\text{C6})$$

This identity fixes the convention and prevents an accidental reversal of a noncommuting word.

If QR is performed once per circuit cycle, the finite-time one-leg growth rates in units of cycles are

$$\ell_a^{(T)}(\xi) = \frac{1}{T} \sum_{t=1}^T \log |(R_t)_{aa}|. \quad (\text{C7})$$

The diagonal phases of R_t are absorbed into Q_t , so the reported diagonal is nonnegative. A rank loss gives an

exact zero and hence $-\infty$; a numerical implementation should report null or censored counts separately rather than silently replacing them by a physical finite rate.

The Oseledets interpretation requires more than the algebra above. For each finite lattice one first takes $T \rightarrow \infty$ along a stationary ergodic Born process, subject to the logarithmic integrability assumptions of the multiplicative ergodic theorem [67]. Only after this temporal limit is controlled should one take the thermodynamic limit. For S independent trajectories the quenched estimator is

$$\overline{\ell_a^{(T)}} = \frac{1}{S} \sum_{\xi=1}^S \frac{1}{T} \sum_t \log |(R_{t,\xi})_{aa}|. \quad (\text{C8})$$

It is neither the logarithm of an averaged product nor the rate obtained from the singular values of $E_\xi \tilde{J}_T$. When the leading physical rates are nonpositive, the corresponding one-leg decay rates are $\gamma_a = -\ell_a$. The covariance derivative is a congruence, so its asymptotic growth rates are the allowed pair sums $\ell_a + \ell_b$, not the individual one-leg values. Every reported “tangent gap” must state which of these two spectra is used. A burn-in interval evolves the base covariance without accumulating R_t . The tangent frame is then initialized, evolved through a declared alignment interval, and retained while its QR log accumulator is reset before the observation window begins.

2. Maximally mixed input and endpoint identities

The maximally mixed covariance is $C_0 = \mathbf{1}/2$, or $G_0 = 0$. For a regular even word with one-particle transfer product M_D , normalized state evolution gives the finite-time identity

$$\tilde{J}_D = 2M_D(\mathbf{1} + M_D^\dagger M_D)^{-1}. \quad (\text{C9})$$

Thus a transfer singular value $s = e^x$ maps to

$$j = \frac{2s}{1 + s^2} = \text{sech } x. \quad (\text{C10})$$

This relation identifies s and s^{-1} ; the tangent response retains the distance from $s = 1$, not its radial orientation. Equation (C9) must not be applied letter by letter to a word containing exact projectors or gain/loss.

There is, however, an exact endpoint statement for an arbitrary nonzero Gaussian word. Let its normalized Choi occupied plane have a reduced frame

$$F_K = \begin{pmatrix} A_K \\ B_K \end{pmatrix}, \quad \mathcal{G}_K = F_K^\dagger F_K. \quad (\text{C11})$$

Then, up to a phase that drops out of the congruence,

$$\boxed{C_D = B_K \mathcal{G}_K^{-1} B_K^\dagger, \quad \tilde{J}_D = 2A_K \mathcal{G}_K^{-1} B_K^\dagger,} \quad (\text{C12})$$

and purity of the Choi projector implies

$$\boxed{\tilde{J}_D^\dagger \tilde{J}_D = 4C_D(\mathbf{1} - C_D).} \quad (\text{C13})$$

The cocycle kernel locates every constrained output direction, while the endpoint covariance labels that direction occupied or empty. This is the precise maximally mixed relation used in the singular theory.

3. Regulated rank-one Choi-covariance composition

The rank loss in Eq. (C3) may also be handled directly at the level of pure Gaussian Choi covariances. Write the incoming two-leg covariance and the local rank-one relation as

$$\Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}, \quad \mathcal{K}(\eta_1, \eta_2) = \begin{pmatrix} \eta_1 P & Q \\ Q & \eta_2 P \end{pmatrix}, \quad (\text{C14})$$

where $P = |\chi\rangle\langle\chi|$, $Q = \mathbf{1} - P$, and $\eta_1, \eta_2 \in \{+1, -1\}$ encode the selected and reset occupations. Let

$$R = \Sigma_{RR}, \quad d = \text{Tr}(PR) - \eta_1. \quad (\text{C15})$$

Proposition (finite regulated contraction). Replace the singular incoming block by the projector-adapted regulator

$$\mathcal{K}_{LL}^{(\epsilon)} = \eta_1 P + \epsilon Q. \quad (\text{C16})$$

If $d \neq 0$, the $\epsilon \rightarrow 0$ contraction exists and is

$$\boxed{\Sigma' = \begin{pmatrix} \Sigma_{LL} - \frac{\Sigma_{LR} P \Sigma_{RL}}{d} & \Sigma_{LR} Q - \frac{\Sigma_{LR} P R Q}{d} \\ Q \Sigma_{RL} - \frac{Q R P \Sigma_{RL}}{d} & \eta_2 P + Q R Q - \frac{Q R P R Q}{d} \end{pmatrix}.} \quad (\text{C17})$$

The regulator acts only on $\text{Ran } Q$ and retains finite terms generated by the divergent complementary inverse. Equation (C17) therefore cannot be obtained by blindly replacing the singular block by its Moore–Penrose pseudoinverse. When $d = 0$, the displayed finite limit does not exist; this

is a zero-overlap or singular selected branch and requires a separate rescaled-limit analysis.

If $\Sigma = \Sigma^\dagger$ and $\Sigma^2 = \mathbf{1}$, then Eq. (C17) obeys

$$\Sigma' = \Sigma'^\dagger, \quad (\Sigma')^2 = \mathbf{1}. \quad (\text{C18})$$

Indeed, choose a basis adapted to $Q \oplus P$ and reorder the covariance as $(L \oplus Q) \oplus P$. Then

$$\tilde{\Sigma} = \begin{pmatrix} X & y \\ y^\dagger & r \end{pmatrix}, \quad \tilde{\Sigma}' = \begin{pmatrix} T & 0 \\ 0 & \eta_2 \end{pmatrix}, \quad T = X - \frac{y y^\dagger}{r - \eta_1}, \quad (\text{C19})$$

with $r = \text{Tr}(PR)$. The original involution gives

$$X^2 + y y^\dagger = \mathbf{1}, \quad X y + r y = 0, \quad y^\dagger y + r^2 = 1. \quad (\text{C20})$$

Using these identities and $\eta_1^2 = 1$,

$$T^2 = \mathbf{1} + \left[-1 + \frac{2r}{d} + \frac{1 - r^2}{d^2} \right] y y^\dagger = \mathbf{1}, \quad (\text{C21})$$

because $-d^2 + 2rd + 1 - r^2 = 1 - \eta_1^2 = 0$; together with $\eta_2^2 = 1$, this proves Eq. (C18).

For completeness, the same pure Choi covariance supplies two complementary transfer charts without assigning an inverse to a projected direction. For equal-dimensional legs write

$$\Sigma = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}. \quad (\text{C22})$$

On their respective finite sectors, the particle and hole charts are

$$\mathcal{T}_p = -(\mathbf{1} + A)^{-1} B, \quad \mathcal{T}_h = (\mathbf{1} - A^\top)^{-1} B^*. \quad (\text{C23})$$

The particle chart has an endpoint pole at $A = -\mathbf{1}$, whereas the hole chart has the complementary pole at $A = +\mathbf{1}$. For every nonzero finite coupled singular value,

$$s_j(\mathcal{T}_h) = s_j(\mathcal{T}_p)^{-1}. \quad (\text{C24})$$

Thus a projected endpoint is covered by the complementary chart together with its occupied or empty cap. Equations (C23) and (C24) are only the chart transition; the quotient-core and radial-response data are given below and are not reconstructed here.

4. Caps, quotient core, and trace anchor

A singular Gaussian word is more naturally a Choi relation than a square transfer matrix. For the reduced frame above, define the occupied plane

$$W_K = \{(A_K v, B_K v) : v \in \mathbb{C}^{r_K}\} \subset \mathcal{H}_L \oplus \mathcal{H}_R. \quad (\text{C25})$$

Its four intrinsic one-leg cap spaces are

$$\begin{aligned} \mathcal{O}_R &= B_K \ker A_K, & \mathcal{E}_R &= (\text{Ran } B_K)^\perp, \\ \mathcal{O}_L &= A_K \ker B_K, & \mathcal{E}_L &= (\text{Ran } A_K)^\perp. \end{aligned} \quad (\text{C26})$$

They are respectively the exact one- and zero-eigenspaces of the diagonal Choi covariance blocks. After these forced directions are removed, the remaining relation is the graph of an invertible quotient map

$$R_{\text{core}} : \mathcal{H}_{\text{in}}^f \longrightarrow \mathcal{H}_{\text{out}}^f, \quad T_{\text{core}} = R_{\text{core}}^\dagger. \quad (\text{C27})$$

The final datum is the trace anchor

$$\nu_K = 2^{-N} \text{Tr}(K^\dagger K), \quad (\text{C28})$$

which is absent from the normalized Choi plane but indispensable for absolute Born probabilities. Caps plus R_{core} determine the normalized endpoint geometry; they do not determine ν_K .

The centered Choi covariance

$$\mathcal{S}_K = 2\mathcal{C}_K - \mathbf{1}, \quad \mathcal{S}_K^\dagger = \mathcal{S}_K, \quad \mathcal{S}_K^2 = \mathbf{1}, \quad (\text{C29})$$

may be read as an auxiliary scattering matrix. On a free singular channel of T_{core} with $s = e^x$, it reduces to

$$\mathcal{S}(x) = \begin{pmatrix} -\tanh x & \text{sech } x \\ \text{sech } x & \tanh x \end{pmatrix}. \quad (\text{C30})$$

The transmission amplitude is the maximally mixed tangent singular value in Eq. (C10). Occupied and empty caps are perfectly reflecting channels with opposite orientation.

This scattering interpretation is auxiliary: $\mathcal{S}_K$ is a coordinate representation of a Choi boundary relation, not the physical time-evolution operator of the fermions. The actual reservoir operation is the local fSWAP in Eq. (A8). Away from maximally mixed input the tangent factors are dressed by the evolving trajectory and cannot be identified with fixed transmission blocks. Likewise, a topological invariant of an auxiliary scatterer is meaningful only after its locality, symmetry, gap, and stability have been established; it is not implied merely by writing Eq. (C29).

Appendix D: Entangling capacity of the local adaptive circuit

We record the preparation bound of Lu, Lessa, Kim, and Hsieh [28] in a form adapted to the present singular instrument. Consider a pure trajectory and a bipartition $\mathcal{H}_A \hat{\otimes} \mathcal{H}_{\bar{A}}$. Let χ_A be its Schmidt rank and $S_0(A) = \log \chi_A$ its Hartley entropy. A branch operator supported entirely within A or $\bar{A}$ cannot increase χ_A . If a crossing operator has operator-Schmidt decomposition

$$O_j = \sum_{\alpha=1}^{r_j} \sigma_{j\alpha} O_{j\alpha}^A \otimes O_{j\alpha}^{\bar{A}}, \quad (\text{D1})$$

then it can multiply the state Schmidt rank by at most r_j . Consequently every nonzero adaptive branch obeys

$$S_0^{\text{out}}(A) - S_0^{\text{in}}(A) \leq \sum_{j \text{ crossing } \partial A} \log r_j. \quad (\text{D2})$$

Normalization by the branch probability is a scalar and does not affect this bound. Neither invertibility nor a lower bound on the Born probability is assumed.

For a depth- D circuit whose layers consist of nonoverlapping, bounded-support gates, the number of crossing gates per layer is $O(|\partial A|)$, while $r_j = O(1)$. Equation (D2) then gives

$$S_0^{\text{out}}(A) - S_0^{\text{in}}(A) \leq O(D|\partial A|). \quad (\text{D3})$$

For a pure product input, $S_0^{\text{in}} = 0$, and every positive-index Rényi entropy satisfies

$$S_n^{\text{out}}(A) \leq S_0^{\text{out}}(A) \leq O(D|\partial A|), \quad n > 0. \quad (\text{D4})$$

The product-input qualification is essential. A Haar-random Slater determinant is already spatially entangled, while the maximally mixed state is not a pure state to which Eq. (D4) directly applies. The latter may be treated by adjoining a colocated purifier, but the cut and the retained reference modes must then be specified.

1. Operator-Schmidt refinement for a corrected orbital

The rank-one fermion instrument permits a sharper count. Across a spatial cut, decompose a_χ in the graded tensor product as

$$a_\chi = a_A \otimes \mathbf{1} + (-1)^{F_A} \otimes a_{\bar{A}}. \quad (\text{D5})$$

It follows that

$$r(a_\chi), r(a_\chi^\dagger) \leq 2, \quad r(n_\chi), r(\mathbf{1} - n_\chi) \leq 4. \quad (\text{D6})$$

Let $N_{\leq}^\partial(\xi)$ count crossing measurements whose result already equals the target, and $N_{\neq}^\partial(\xi)$ count crossing measurements that are corrected by gain or loss. Equations (A6) and (D2) give the branchwise bound

$$S_0^\xi(A, T) - S_0(A, 0) \leq 2 \log 2 N_{\leq}^\partial(\xi) + \log 2 N_{\neq}^\partial(\xi). \quad (\text{D7})$$

This is a useful implementation check because every quantity on the right-hand side is available from the same trajectory-resolved record.

2. Full-width strip and the two-dimensional resource

The entropy used to diagnose the wall is computed for the full-width strip

$$A_\ell = \{(x, y) : 0 \leq x < N_x, 0 \leq y < \ell\}. \quad (\text{D8})$$

On a torus, ∂A_ℓ contains two transverse cuts of length N_x . Thus $|\partial A_\ell| = O(N_x)$, not $O(1)$. A subsequent restriction of the entanglement contour to a window in x does not change the subsystem and is not itself the entropy of a wall-only region.

Singular Choi geometry and normalized response

Panel (a): the four occupied/empty cap spaces and the finite transmitted quotient core. Panel (b): radial free-channel block, showing that $s \leftrightarrow s^{-1}$ reverses reflection orientation but leaves the tangent transmission $2s/(1+s^2)$ unchanged. Panel (c): the chronological state-dressed tangent product and QR accumulation. The caption should state explicitly that the Choi scatterer is auxiliary and that the physical source/drain primitive is fSWAP with a fresh local ancilla.

FIG. 8. Singular Choi geometry and normalized response. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

Suppose four rank-one controller orbitals are applied at every active center, their square support has radius r , and the active slab has transverse width W . Away from the short-interval regime in which the two cuts overlap, at most $2r$ center rows cross each cut. Hence one complete sweep contains at most

$$N_{\partial}(T) \leq 16rWT \quad (\text{D9})$$

crossing letters after T sweeps. From a pure product input,

$$S_n(A_\ell, T) \leq 32rWT \log 2. \quad (\text{D10})$$

For two oppositely oriented complex-fermion walls, the proposed CFT form is

$$S_1(A_\ell) = b(N_x) + \frac{c_L + c_R}{6} \log \left[\frac{N_y}{\pi a} \sin \left(\frac{\pi \ell}{N_y} \right) \right] + \dots, \quad (\text{D11})$$

with $c_L = c_R = 1$ in the ideal benchmark. If this form is established from a pure product input, Eqs. (D10) and (D11) imply the asymptotic resource tradeoff

$$WT = \Omega(\log N_y). \quad (\text{D12})$$

The intrinsic one-dimensional result $D = \Omega(\log N_y)$ is recovered when the transverse resource is fixed. In a two-dimensional aspect-ratio limit, the growing transverse boundary supplies entangling capacity.

3. Raster sweeps and finite-color schedules

The variable T in Eq. (D9) counts complete controller sweeps; it is not automatically the circuit depth D in Eq. (D3). A raster sweep applies overlapping OW projectors in a prescribed serial order. Because neighboring projectors are generically noncommuting, reproducing that precise word has an extensive causal depth.

For $n_{\text{shell}} = 1$, let $u = (x, y)$ denote an update center and write its cell support as

$$\mathcal{S}_u = \{u + (\delta_x, \delta_y) : \delta_x, \delta_y \in \{-1, 0, 1\}\}, \quad (\text{D13})$$

with both coordinates understood periodically. The support-conflict graph $\mathcal{G}_{\text{conf}} = (\mathcal{V}, \mathcal{E})$ has one vertex for each active center and an edge $(u, v) \in \mathcal{E}$ precisely when $\mathcal{S}_u \cap \mathcal{S}_v \neq \emptyset$. In terms of the cyclic distance

$$d_L(a, b) = \min_{m \in \mathbb{Z}} |a - b + mL|, \quad (\text{D14})$$

two distinct centers conflict if and only if

$$d_{N_x}(x, x') \leq 2 \quad \text{and} \quad d_{N_y}(y, y') \leq 2. \quad (\text{D15})$$

The degree is therefore at most 24, so a finite coloring with at most 25 colors exists even without using the translational structure.

On a torus for which $q_x \mid N_x$, $q_y \mid N_y$, and $q_x, q_y \geq 3$, the residue map

$$c(x, y) = (x \bmod q_x, y \bmod q_y) \quad (\text{D16})$$

is a proper $q_x q_y$ -coloring. Indeed, two distinct centers of the same color differ by a nonzero multiple of q_x or q_y in at least one coordinate. Their cyclic separation in that coordinate is at least three, so Eq. (D15) fails and the supports are disjoint. Thus a commensurate 3×3 coloring gives nine colors. For the proposed sequence

$$N_x = 20, \quad N_y \in \{24, 32, 40, 48, 64, 80\}, \quad (\text{D17})$$

the conservative choice $q_x = q_y = 4$ wraps periodically and gives

$$\chi_{\text{conf}} \leq 16. \quad (\text{D18})$$

All centers of a fixed color can be updated in parallel. If the four controller orbitals are kept as separate instrument blocks and one occupation-measurement-feedforward-fSWAP block has bounded physical depth $d_{\text{inst}} = O(1)$, a complete colored sweep obeys

$$d_{\text{cyc}} \leq 4\chi_{\text{conf}} d_{\text{inst}} \leq 64d_{\text{inst}}. \quad (\text{D19})$$

The bound is independent of N_x and N_y . Consequently $D = d_{\text{cyc}} T_{\text{cyc}}$, and the two-wall entropy $S_1 \sim \frac{1}{3} \log N_y$, together with Eq. (D3), implies

$$T_{\text{cyc}} \gtrsim \frac{\log N_y}{3\kappa d_{\text{cyc}} N_x} \geq \frac{\log N_y}{192\kappa d_{\text{inst}} N_x}, \quad (\text{D20})$$

where κ absorbs the local operator-Schmidt constants and the two components of ∂A_ℓ . At fixed N_x , this rules out a sublogarithmic number of colored cycles for preparation from a product state, but its unknown prefactor is far too weak to prescribe a practical burn-in. The constant bound in Eq. (D19) is per colored sweep only. It does not make the total depth D size independent, and the BPJ result on rapid bulk preparation cannot be transferred without proof to the additional logarithmically entangled wall sector.

This implementation is not a reinterpretation of raster data. If u and v overlap, their conditional instruments generally satisfy

$$\mathcal{M}_u \circ \mathcal{M}_v \neq \mathcal{M}_v \circ \mathcal{M}_u. \quad (\text{D21})$$

Changing from the raster word to color blocks therefore defines a different microscopic circuit, even though both use the same local controller. An orientation-unbiased implementation may draw a fresh permutation of the color blocks each cycle, updating all sites within one block in parallel; the color order and its random seed must be recorded. Raster and reverse-raster schedules remain useful artifact controls. The colored protocol can replace the serial protocol in production only after topology, entanglement and charge coefficients, chirality, and the tangent slow sector agree within predeclared finite-size confidence intervals.

4. Finite wall separation and hybridization

On a periodic transverse direction there are two oppositely oriented walls. If their minimum separation is $w_{\min}$, exponentially localized edge modes generically acquire a hybridization mass

$$m(w_{\min}) \simeq m_0 e^{-w_{\min}/\xi_\perp}. \quad (\text{D22})$$

The corresponding low-energy two-wall Hamiltonian has the Dirac form

$$h_{\text{edge}}(k_y) = vk_y \tau_z + m(w_{\min}) \tau_x. \quad (\text{D23})$$

Critical finite-size scaling on a circumference N_y requires the hybridization scale to remain below the level spacing,

$$m(w_{\min}) \ll \frac{2\pi v}{N_y}. \quad (\text{D24})$$

Therefore an effectively decoupled-wall sequence should satisfy

$$w_{\min} \gtrsim \xi_\perp \log\left(\frac{m_0 N_y}{2\pi v}\right). \quad (\text{D25})$$

This physical width requirement mirrors Eq. (D12). A controlled test should extract $\xi_\perp$ independently from the transverse wall profile and seek a collapse in the variable $N_y e^{-w_{\min}/\xi_\perp}$.

Appendix E: Exact-Hamiltonian benchmark and boundary twists

The equilibrium reference is the two-band class-A lattice Hamiltonian

$$h(\mathbf{k}; \alpha) = \sin k_x \sigma_x + \sin k_y \sigma_y + (\alpha - \cos k_x - \cos k_y) \sigma_z. \quad (\text{E1})$$

A spatial profile $\alpha(x)$ is Fourier transformed only in the translation-invariant direction, or assembled directly as a real-space Hermitian matrix H_{ex} on $2N_x N_y$ orbitals. The exact benchmark contains no Born sampling and no feedback.

At half filling, diagonalize

$$H_{\text{ex}} V = V \text{diag}(\varepsilon_1, \dots, \varepsilon_{2N_x N_y}), \quad \varepsilon_1 \leq \dots \leq \varepsilon_{2N_x N_y}, \quad (\text{E2})$$

and occupy exactly $N_x N_y$ orbitals,

$$P_{\text{ex}} = V_{\text{occ}} V_{\text{occ}}^\dagger, \quad C_{\text{ex}} = P_{\text{ex}}^*, \\ G_{\text{ex}} = 2C_{\text{ex}} - \mathbf{1}, \quad V_{\text{occ}} = (v_1, \dots, v_{N_x N_y}). \quad (\text{E3})$$

Here P_{ex} is the ket-space spectral projector, whereas C_{ex} is the matrix stored with the convention of Eq. (A2). Topological estimators act on $P_{\text{ex}} = C_{\text{ex}}^\top$; covariance observables act on C_{ex} . Selecting states merely by the sign of ε is ambiguous when a finite-size wall level sits at zero. Fixed particle number is the primary half-filling convention. Any residual degeneracy at the Fermi level must be resolved by a stated symmetry, an infinitesimal twist, or explicit subspace tracking.

1. Physical and modular current densities

For a general number-conserving quadratic Hamiltonian, introduce the orbital spinor $\Psi_{\mathbf{r}}$ in cell $\mathbf{r}$ and write

$$H = \sum_{\mathbf{r}, \mathbf{r}'} \Psi_{\mathbf{r}}^\dagger h_{\mathbf{r}\mathbf{r}'} \Psi_{\mathbf{r}'}, \quad n_{\mathbf{r}} = \Psi_{\mathbf{r}}^\dagger \Psi_{\mathbf{r}}. \quad (\text{E4})$$

Hermiticity requires $h_{\mathbf{r}\mathbf{r}'} = h_{\mathbf{r}'\mathbf{r}}^\dagger$. The Heisenberg equation gives the lattice continuity equation

$$\dot{n}_{\mathbf{r}} + \sum_{\mathbf{r}'} J_{\mathbf{r} \rightarrow \mathbf{r}'} = 0, \quad (\text{E5})$$

with the positively oriented, outward bond-current operator

$$J_{\mathbf{r} \rightarrow \mathbf{r}'} = \mathbf{i} \Psi_{\mathbf{r}}^\dagger h_{\mathbf{r}\mathbf{r}'} \Psi_{\mathbf{r}'} - \mathbf{i} \Psi_{\mathbf{r}'}^\dagger h_{\mathbf{r}'\mathbf{r}} \Psi_{\mathbf{r}}. \quad (\text{E6})$$

Using the covariance convention of Eq. (A2), its Gaussian expectation value is

$$\langle j_{\mathbf{r} \rightarrow \mathbf{r}'} \rangle = 2 \text{Im Tr}[h_{\mathbf{r}\mathbf{r}'} C_{\mathbf{r}'\mathbf{r}}], \quad j_{\mathbf{r} \rightarrow \mathbf{r}'} = -j_{\mathbf{r}' \rightarrow \mathbf{r}}. \quad (\text{E7})$$

For the exact QWZ benchmark of Eq. (E1), the two positively oriented nearest-neighbor currents are therefore

$$j_x(\mathbf{r}) = 2 \text{Im Tr}[h_{\mathbf{r}, \mathbf{r}+\hat{x}} C_{\mathbf{r}+\hat{x}, \mathbf{r}}], \\ j_y(\mathbf{r}) = 2 \text{Im Tr}[h_{\mathbf{r}, \mathbf{r}+\hat{y}} C_{\mathbf{r}+\hat{y}, \mathbf{r}}]. \quad (\text{E8})$$

Where the preparation resource resides

Panel (a): a full-width strip with its two transverse entanglement cuts and two domain walls. Panel (b): crossing branch operators counted in Eq. (D7). Panel (c): raster versus finite-color scheduling. Panel (d): the two-wall hybridization crossover controlled by $N_y e^{-w_{\min}/\xi_{\perp}}$. All plotted curves and fit parameters are placeholders until the corresponding campaigns are complete.

FIG. 9. Where the preparation resource resides. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

The same construction defines a modular current for a subsystem A . On the entanglement-active support $0 < C_A < \mathbf{1}_A$, the one-body modular Hamiltonian is

$$\begin{aligned} h_A^{\text{mod}} &= \log[(\mathbf{1}_A - C_A)C_A^{-1}] \\ &= -2 \operatorname{arctanh}(G_A), \quad G_A = 2C_A - \mathbf{1}_A. \end{aligned} \quad (\text{E9})$$

Replacing $h_{\mathbf{r}\mathbf{r}'}$ in Eq. (E7) by the corresponding blocks of h_A^{mod} , while retaining C_A , gives the modular-current map. Exact eigenvalues 0 or 1 lie outside this finite chart. Numerically they are either removed by restricting to the entanglement-active support or clipped to $[\epsilon, 1 - \epsilon]$; any spatial current claim must be stable as ϵ is reduced.

Equations (E5)–(E8) are source free for exact-Hamiltonian and modular evolution. The adaptive system alone exchanges charge with measured and fSWAP ancillas, so a circuit-time balance law contains a local source term,

$$\Delta n_{\mathbf{r}} + (\nabla_{\text{lat}} \cdot j)_{\mathbf{r}} = s_{\mathbf{r}}^{\text{anc}}. \quad (\text{E10})$$

Consequently a source-free physical current is not inferred from the measurement/feedforward occupancy change; charge conservation holds instead on the enlarged system–ancilla Hilbert space.

2. Twisted boundary condition

A twist ϕ_y is inserted into the same Hamiltonian by multiplying the forward hopping across one y -seam by $e^{i\phi_y}$ and the reverse hopping by $e^{-i\phi_y}$. A uniform gauge distributes the phase as $e^{i\phi_y/N_y}$ over all forward y -links and is unitarily equivalent. The spatial profile, disorder realization, and every other parameter are held fixed as ϕ_y varies.

Let $V_j = V_{\text{occ}}(\phi_j)$ be an orthonormal occupied frame. Neighboring frames are aligned through the singular-value decomposition

$$M_j = V_j^\dagger V_{j+1} = L_j \Sigma_j R_j^\dagger, \quad \mathcal{U}_j = L_j R_j^\dagger. \quad (\text{E11})$$

Multiplying V_{j+1} by $R_j L_j^\dagger$ makes the overlap with V_j positive and fixes a discrete parallel-transport gauge. The Wilson holonomy and its many-body phase are

$$\mathcal{W}_y = \mathcal{U}_0 \mathcal{U}_1 \cdots \mathcal{U}_{N_\phi-1}, \quad \gamma_y = \arg \det \mathcal{W}_y. \quad (\text{E12})$$

This polar alignment is a gauge choice for one continuous family. It cannot turn independently sampled Hamiltonians or independently sampled circuit records into such a family.

If the half-filled gap remains open, $P_{\text{ex}}(\phi_y)$ defines a smooth ground-state bundle. At an exact spectral crossing, there are two distinct questions: the instantaneous half-filled ground-state projector may switch discontinuously, whereas a maximally overlapping tracked subspace follows a spectral-flow branch and need not remain the instantaneous ground state. Both are useful, but they must be labeled separately.

For a two-twist grid $\phi = (\phi_x, \phi_y)$, a gauge-invariant lattice Chern number is obtained from determinant links

$$U_\mu(\phi) = \frac{\det[V^\dagger(\phi)V(\phi + \hat{\mu})]}{|\det[V^\dagger(\phi)V(\phi + \hat{\mu})]|}, \quad (\text{E13})$$

and plaquette curvature

$$\begin{aligned} F_{xy}(\phi) &= \text{Arg}[U_x(\phi)U_y(\phi + \hat{x})U_x(\phi + \hat{y})^{-1}U_y(\phi)^{-1}], \\ C_{\text{tw}} &= \frac{1}{2\pi} \sum_{\phi} F_{xy}. \end{aligned} \quad (\text{E14})$$

The mesh must be refined until the admissibility condition and the integer are stable [58].

The exact benchmark comparison should report separately: the half-filled spectrum and wall velocity, the transverse localization length, the strip entropy and contour, the local or twisted-boundary Chern diagnostic, and the finite-width hybridization gap. Agreement of a circuit observable with one benchmark does not imply equality of the two states.

Exact Hamiltonian, half filling, and twist tracking

Panel (a): the real-space domain-wall profile and exact low-energy spectrum. Panel (b): occupied-subspace tracking under a boundary twist, including the distinction between instantaneous ground-state filling and a spectral-flow branch. Panel (c): convergence of the twist-mesh Chern estimator. Panel (d): proposed circuit-versus-exact comparisons. Numerical values are left blank until established.

FIG. 10. Exact Hamiltonian, half filling, and twist tracking. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

Appendix F: Trajectory estimators and averaging conventions

Every stochastic estimator below is evaluated on independently Born-sampled records. Equal sample weights therefore estimate the physical Born ensemble; no second factor of p_ξ is inserted. We use an overbar for this Monte Carlo average,

$$\overline{X_\xi} = \frac{1}{S} \sum_{\xi=1}^S X_\xi. \quad (\text{F1})$$

Nonlinear functions are evaluated trajectory by trajectory unless an explicitly annealed quantity is stated.

1. Entropy and entanglement contour

For a Gaussian trajectory and a spatial subsystem A , let $C_{\xi,A}$ be the restricted two-point matrix. The Gaussian von Neumann entropy is

$$S_\xi(A) = -\text{tr}[C_{\xi,A} \log C_{\xi,A}] - \text{tr}[(\mathbf{1}_A - C_{\xi,A}) \log(\mathbf{1}_A - C_{\xi,A})]. \quad (\text{F2})$$

If $C_{\xi,A} u_{\xi,\alpha} = \nu_{\xi,\alpha} u_{\xi,\alpha}$, define $h(\nu) = -\nu \log \nu - (1 - \nu) \log(1 - \nu)$. A site-resolved contour is

$$s_{\xi,A}(i) = \sum_{\alpha} |u_{\xi,\alpha}(i)|^2 h(\nu_{\xi,\alpha}) = [h(C_{\xi,A})]_{ii}. \quad (\text{F3})$$

Internal orbitals in one unit cell are summed only after Eq. (F3). It follows exactly that

$$S_\xi(A) = \sum_{i \in A} s_{\xi,A}(i). \quad (\text{F4})$$

For a mixed trajectory, Eq. (F2) is the entropy of the reduced mixed state, not an entanglement measure. Purity, or an appropriate mixed-state entanglement diagnostic, must accompany any such interpretation.

For the full-width strip A_ℓ , a wall-window contour sum is

$$\mathcal{S}_{\xi,I_x}(\ell) = \sum_{x \in I_x} \sum_{y=0}^{\ell-1} \sum_{\mu} s_{\xi,A_\ell}(x, y, \mu). \quad (\text{F5})$$

Only $I_x = [0, N_x - 1]$ gives the entropy of A_ℓ . A narrow I_x is a localization diagnostic for the contour of the same full-width subsystem; it is not the entropy of a narrow tube.

The primary trajectory estimator is

$$\overline{S(A_\ell)} = \frac{1}{S} \sum_{\xi} S_\xi(A_\ell), \quad (\text{F6})$$

which generally differs from both $S[\overline{C_A}]$ and the entropy of the averaged many-body density matrix. A proposed two-wall CFT fit uses

$$d_{N_y}(\ell) = \frac{N_y}{\pi} \sin\left(\frac{\pi \ell}{N_y}\right),$$

$$\overline{S(A_\ell)} = b + \frac{c_L + c_R}{6} \log d_{N_y}(\ell) + \dots \quad (\text{F7})$$

Fit windows, endpoint exclusions, finite-size corrections, and covariance of the fitted points must be specified. A coefficient inferred from $\mathcal{S}_{I_x}$ for one wall is supporting contour evidence, not a standalone entropy measurement.

2. Conditional mutual information and the tripartite reduction

Let A, B, C be adjacent intervals in the periodic wall direction, with each region extending across the same full transverse width. For every trajectory ξ , define the mutual information between the separated intervals and the standard tripartite information by

$$I_{2,\xi}(A, C) = S_\xi(A) + S_\xi(C) - S_\xi(A \cup C),$$

$$I_{3,\xi}(A, B, C) = S_\xi(A) + S_\xi(B) + S_\xi(C) - S_\xi(A \cup B) - S_\xi(A \cup C) - S_\xi(B \cup C) + S_\xi(A \cup B \cup C). \quad (\text{F8})$$

Their difference eliminates the disconnected-region entropy exactly:

$$\begin{aligned}\mathcal{I}_\xi(A : C|B) &\equiv I_{2,\xi}(A, C) - I_{3,\xi}(A, B, C) \\ &= S_\xi(A \cup B) + S_\xi(B \cup C) \\ &\quad - S_\xi(B) - S_\xi(A \cup B \cup C) \\ &= I_\xi(A : C|B),\end{aligned}\quad (\text{F9})$$

so the quantity called the “tripartite reduction” is precisely a conditional mutual information and is nonnegative by strong subadditivity.

For a common interval family governed by a one-dimensional CFT, the averaged entropy has the finite-ring form [77]

$$\overline{S(X)} = \kappa \log d_L(\ell_X) + s_0 + \dots, \quad d_L(\ell) = \frac{L}{\pi} \sin\left(\frac{\pi\ell}{L}\right), \quad (\text{F10})$$

where κ includes the number and orientation-independent content of the wall branches intersected by the chosen geometry. Substitution into Eq. (F9) cancels s_0 and gives

$$\overline{\mathcal{I}(A : C|B)} = \kappa \log \left[\frac{d_L(A \cup B)d_L(B \cup C)}{d_L(B)d_L(A \cup B \cup C)} \right] + \dots. \quad (\text{F11})$$

Define the chord cross ratio

$$x = \frac{d_L(A)d_L(C)}{d_L(A \cup B)d_L(B \cup C)}. \quad (\text{F12})$$

For adjacent lengths ℓ_A, ℓ_B, ℓ_C , the elementary identity

$$\sin(a+b)\sin(b+c) - \sin a \sin c = \sin b \sin(a+b+c) \quad (\text{F13})$$

with $a = \pi\ell_A/L$, $b = \pi\ell_B/L$, and $c = \pi\ell_C/L$ implies

$$d_L(A \cup B)d_L(B \cup C) - d_L(A)d_L(C) = d_L(B)d_L(A \cup B \cup C). \quad (\text{F14})$$

Consequently,

$$\overline{I_2 - I_3} = \kappa \log \left(\frac{1}{1-x} \right) + \dots. \quad (\text{F15})$$

The estimator order is part of this statement. Each of the four entropies in Eq. (F9) is computed from the same trajectory covariance before forming $\mathcal{I}_\xi$, and only then is the equal-weight Born average taken:

$$\overline{\mathcal{I}(A : C|B)} = \frac{1}{S} \sum_{\xi=1}^S \mathcal{I}_\xi(A : C|B). \quad (\text{F16})$$

Using $S[\overline{C}]$, or mixing entropies drawn from independently sampled records, defines a different quantity. For a genuine single complex chiral wall with $c_L + c_R = 1$, Eq. (F15) has $\kappa = 1/6$. A full-width strip intersecting the two oppositely oriented walls has $\kappa = 1/3$. This factor of two counts wall branches and does not measure their propagation direction.

TABLE I. Exact-Hamiltonian calibration of $I_2 - I_3$ against $\log[1/(1-x)]$. Absolute fit uncertainties are displayed explicitly.

Estimator	Fitted slope [CFT]	R^2
Each wall-window	0.166967668	0.999999945
contour sum	± 0.000027005 [1/6]	
Full-width CMI	0.334232464	0.99999829
	± 0.000095377 [1/3]	

The archived exact-domain-wall ground state provides a deterministic calibration of this estimator at $N_x \times N_y = 20 \times 40$, with $\ell_A = \ell_C = 3$. The fitted slopes are summarized in Table I. Only the full-width row is a conditional mutual information of spatial subsystems. In the first row the subsystem is still defined over the full transverse width and only the final entanglement-contour sum is restricted to one wall; those values test localization and the factor of two, not a standalone one-wall CMI. Both rows are deterministic exact-Hamiltonian calibrations. They do not constitute a production Born-trajectory result for the adaptive circuit, and Eq. (F15) is not an independent central-charge estimator when it is assembled from the same entropies used in Eq. (F7).

3. Quantum charge variance, full counting statistics, and current algebra

For $N_A = \sum_{i \in A} c_i^\dagger c_i$, the charge expectation and the within-trajectory quantum variance are

$$\begin{aligned}\langle N_A \rangle_\xi &= \text{tr } C_{\xi,A}, \\ F_{\xi,A}^{(q)} &= \langle (N_A - \langle N_A \rangle_\xi)^2 \rangle_\xi \\ &= \text{tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})].\end{aligned}\quad (\text{F17})$$

The total variance obtained by first drawing a Born trajectory and then measuring N_A obeys the law of total variance,

$$F_A^{\text{total}} = \overline{F_{\xi,A}^{(q)}} + \text{Var}_\xi(\text{tr } C_{\xi,A}). \quad (\text{F18})$$

The second term is classical record-to-record charge noise. It must not be included in a Kac–Moody coefficient intended to characterize quantum charge fluctuations of the conditional state.

The trajectory full-counting-statistics generator is the determinant

$$\mathcal{Z}_{\xi,A}(\lambda) = \langle e^{i\lambda N_A} \rangle_\xi = \det[\mathbf{1}_A - C_{\xi,A} + e^{i\lambda} C_{\xi,A}], \quad (\text{F19})$$

with cumulants

$$\kappa_{n,\xi}^{(q)} = (-i\partial_\lambda)^n \log \mathcal{Z}_{\xi,A}(\lambda) \big|_{\lambda=0}. \quad (\text{F20})$$

The Born-ensemble characteristic function $\overline{\mathcal{Z}_{\xi,A}}$ describes the two-stage physical sampling experiment. The typical generator $\exp[\log \mathcal{Z}_{\xi,A}]$ is a different, quenched quantity.

With current normalization

$$\langle J(z)J(0) \rangle = \frac{k}{z^2}, \quad (\text{F21})$$

a single chiral branch at level k contributes

$$\overline{F_{\xi, A_\ell}^{(q)}} = b_F + \frac{k}{2\pi^2} \log d_{N_y}(\ell) + \dots \quad (\text{F22})$$

For two independent opposite walls, the coefficients add:

$$\overline{F_{\xi, A_\ell}^{(q)}} = b_F + \frac{k_L + k_R}{2\pi^2} \log d_{N_y}(\ell) + \dots \quad (\text{F23})$$

The ideal complex-fermion benchmark has $k_L = k_R = 1$. In that free benchmark the leading terms satisfy

$$S_{\text{wall}} = \frac{\pi^2}{3} F_{\text{wall}}^{(q)} + \dots \quad (\text{F24})$$

The paper should treat Eqs. (F23) and (F24) as tests to be performed, rather than as consequences of the observed entropy coefficient.

4. Correlations and spatial mode densities

Wick's theorem gives the connected density correlator on one trajectory,

$$\langle \delta n_i \delta n_j \rangle_\xi = \delta_{ij} C_{\xi, ii} - |C_{\xi, ij}|^2. \quad (\text{F25})$$

For separated cells one may therefore use the positive squared-correlation diagnostic

$$C_\xi(x, r) = \frac{1}{N_y} \sum_y \sum_{\mu, \nu} |C_{\xi, (x, y, \mu), (x, y+r, \nu)}|^2. \quad (\text{F26})$$

The annealed and typical correlations are respectively

$$C_{\text{ann}}(x, r) = \overline{C_\xi(x, r)}, \quad C_{\text{typ}}(x, r) = \exp \left[\overline{\log C_\xi(x, r)} \right]. \quad (\text{F27})$$

They need not share an exponent in a broad stochastic ensemble. In particular,

$$\overline{|C_{\xi, ij}|^2} \neq |\overline{C_{\xi, ij}}|^2. \quad (\text{F28})$$

Every reported log-log fit must state whether it uses $\log \overline{C_\xi}$ or $\log \overline{C_\xi}$.

For a normalized tangent or endpoint vector $v_{\xi, a}$, the physical component-summed spatial density is

$$\rho_a(x, y) = \frac{1}{S} \sum_{\xi=1}^S \sum_{\mu} |v_{\xi, a}(x, y, \mu)|^2, \quad \sum_{x, y} \rho_a(x, y) = 1. \quad (\text{F29})$$

Normalizing each internal component μ separately before averaging instead produces the spatial distribution conditioned on that component and discards its physical relative weight. Such a visualization, if used, must be labeled accordingly.

Statistical uncertainty is estimated over independent trajectories, not over spatial points reused within one trajectory. Translation averages reduce variance but do not create additional independent samples. Bootstrap or jackknife resampling should preserve the full trajectory as the elementary block.

Appendix G: Born-ensemble response and fixed-record holonomy

1. The score term in a physical response

Let a perturbation parameter u modify the microscopic circuit, and let $O_\xi(u)$ be an observable on the normalized trajectory selected by ξ . The Born-ensemble expectation is

$$\langle O \rangle_B(u) = \sum_{\xi} p_\xi(u) O_\xi(u). \quad (\text{G1})$$

Differentiating gives the exact likelihood-ratio identity

$$\partial_u \langle O \rangle_B = \overline{\partial_u O_\xi} + \overline{O_\xi \partial_u \log p_\xi}. \quad (\text{G2})$$

Since $\overline{\partial_u \log p_\xi} = 0$, the second term may equivalently be written as a covariance with the score. The first term is the fixed-record conditional response; the second accounts for the change in the probability of drawing that record. Dropping it does not give the linear response of the physical Born ensemble.

The score is accumulated chronologically,

$$\partial_u \log p_\xi = \sum_t \partial_u \log p_{m_t}(G_{t-1}(u), \chi_t(u)), \quad (\text{G3})$$

where the derivative is total and includes the perturbed earlier covariance. For the binary measurement of Eq. (A5),

$$\partial_u \log p_{m_t} = \frac{\eta_{m_t} \partial_u g_{\chi_t}}{1 + \eta_{m_t} g_{\chi_t}}, \quad (\text{G4})$$

with

$$\partial_u g_\chi = \chi^\dagger (\partial_u G) \chi + 2 \text{Re} [(\partial_u \chi)^\dagger G \chi]. \quad (\text{G5})$$

The tangent recursion supplies $\partial_u G$, while the second term is retained when the perturbation changes the measured orbital itself.

For a wall-density response to a localized potential,

$$\chi_B(x, t) = \left. \frac{\delta \overline{\langle n(x, t) \rangle_\xi}}{\delta V(0, 0)} \right|_{V=0}, \quad (\text{G6})$$

Eq. (G2) fixes the estimator. Fourier transformation along the wall may then be used to search for a low-energy ridge in $\chi_B(k, \omega)$. A one-sided slope and its reversal under wall-orientation reversal would diagnose chirality; no quantized transport coefficient follows from the ridge alone.

2. Fixed-record twist family

For a circuit record ξ , insert a twist ϕ into the same finite-support orbitals and fSWAP couplings while holding the sequence of outcome labels fixed:

$$K_\xi(\phi) = K_{m_D|s_D}(\phi) \cdots K_{m_1|s_1}(\phi). \quad (\text{G7})$$

The normalized state

$$\rho_\xi(\phi) = \frac{K_\xi(\phi)\rho_0 K_\xi(\phi)^\dagger}{p_\xi(\phi)} \quad (\text{G8})$$

defines a continuous family only on intervals where $p_\xi(\phi) > 0$ and the relevant occupied rank is constant. A zero of the branch probability is a genuine singularity of this normalized family.

For a pure Gaussian branch with occupied frame $V_{\xi,j}$ on a twist mesh, use the polar links of Eq. (E11) and form

$$\mathcal{W}_\xi = \prod_j \text{polar}\left(V_{\xi,j}^\dagger V_{\xi,j+1}\right), \quad \gamma_\xi = \arg \det \mathcal{W}_\xi. \quad (\text{G9})$$

For a mixed Gaussian state, the appropriate object is instead a specified purification, Uhlmann holonomy, or a spectral projector of the modular Hamiltonian; these choices are not interchangeable.

Independently resampling a Hamiltonian, record, or ground state at every ϕ does not define Berry transport. SVD alignment can fix the gauge within one family, but it cannot manufacture continuity between unrelated samples. A single $0 \leq \phi_y \leq 2\pi$ loop measures a holonomy or spectral flow. A Chern number requires a two-dimensional twist family, such as Eq. (E14), or an independently justified real-space invariant.

Appendix H: Conditioned trajectories, mean channel, and record weights

For one adaptive step, denote the unnormalized branch map by

$$\mathcal{K}_{m|s}(\rho) = K_{m|s}\rho K_{m|s}^\dagger. \quad (\text{H1})$$

There are three distinct objects:

$$\rho_{m|s} = \frac{\mathcal{K}_{m|s}(\rho)}{\text{Tr } \mathcal{K}_{m|s}(\rho)} \quad \text{conditioned trajectory,} \quad (\text{H2})$$

$$\mathcal{E}_s(\rho) = \sum_m \mathcal{K}_{m|s}(\rho) \quad \text{unconditional channel,} \quad (\text{H3})$$

$$\rho_{m_*} = \frac{\Pi_{m_*}\rho\Pi_{m_*}}{\text{Tr}(\rho\Pi_{m_*})} \quad \text{postselected outcome.} \quad (\text{H4})$$

The protocol uses the first object with Born-distributed m and deterministic feedback. A no-click non-Hermitian Hamiltonian or forced measurement retains only a selected branch and is not the same instrument.

The one-step perfect-correction average is the Gaussian replacement channel in Eq. (A16). More general convex mixtures of Gaussian Kraus maps need not preserve Gaussianity, especially when future operations depend on coarse-grained records. In either case the conceptual distinction remains:

$$\overline{S(C'_\xi)} \neq S(\overline{C}), \quad |\overline{C'_{\xi,ij}}|^2 \neq |\overline{C_{ij}}|^2. \quad (\text{H5})$$

Even when $\bar{\rho}$ happens to be Gaussian, its entropy is the entropy of the mixed mean state, not the average trajectory entanglement. The tangent cocycle is the derivative of the normalized conditioned map and is not the spectrum of $\mathcal{E}_s$.

The full trajectory-resolved record also defines classical information-theoretic observables. With

$$\mathcal{Z}_\xi = p_\xi, \quad \mathcal{I}_\xi = -\log p_\xi, \quad (\text{H6})$$

the Shannon record entropy and the replicated sums are

$$H_{\text{rec}} = -\sum_\xi p_\xi \log p_\xi = \overline{\mathcal{I}_\xi},$$

$$\mathcal{Z}_n = \sum_\xi p_\xi^n, \quad H_n^{\text{rec}} = \frac{\log \mathcal{Z}_n}{1-n}. \quad (\text{H7})$$

Here $\overline{\mathcal{I}_\xi}$ is exactly the Shannon self-information. Naming it a quenched free energy does not by itself supply a statistical-mechanics theory. The required bridge is

$$-\partial_n \log \mathcal{Z}_n|_{n=1} = H_{\text{rec}}, \quad (\text{H8})$$

together with an explicit replicated transfer normalization and boundary conditions for $\mathcal{Z}_n$. Extracting a central charge or operator content from finite-size corrections then requires an independent record velocity, aspect-ratio convergence, and charge-resolved transfer sectors, following, for example, Ref. [55]; it is not implied by the state entanglement fit.

Appendix I: Robustness protocol and numerical controls

The numerical claims in the main text should be supported by a common set of controls. The following list defines the intended tests; no outcome is assumed here.

Initial states and convergence time. In addition to a generic pure Gaussian state and the maximally mixed state, use onsite Fock product states

$$G_{\text{prod}} = \text{diag}(2n_1-1, \dots, 2n_N-1), \quad n_i \in \{0, 1\}. \quad (\text{I1})$$

Vacuum, deterministic half-filled patterns, and random onsite occupations test distinct charge sectors without importing initial spatial entanglement. For each observable X , define a tolerance-based convergence time

$$T_\epsilon(X) = \min\left\{T : \max_{T' \in [T, T+\Delta T]} |X(T') - X_{\text{late}}| < \epsilon\right\}. \quad (\text{I2})$$

Its scaling with N_x, N_y must be measured rather than fixed by choosing the final cycle proportional to system size.

Locality and support. Repeat the analysis for several finite n_{shell} , and compare with the dense frame only as a nonlocal reference. A trend with support radius must be separated from a trend with system size. Compare the explicit interface and support-terminated protocols as distinct microscopic families.

Schedule. Compare raster, randomized, and finite-color schedules at matched numbers of local letters and at matched physical layer depth. Record both the gate count and the nonoverlapping circuit depth. Universality under schedule changes is stronger evidence than agreement after merely relabeling one sweep as one unit of time.

Imperfect feedback and ancilla noise. Let ϵ_a be the probability that the fresh ancilla is prepared with the wrong occupation, and let p_g and p_l be the success probabilities of gain and loss. The ideal point is

$$\epsilon_a = 0, \quad p_g = p_l = 1. \quad (\text{I3})$$

Vary these independently, because a symmetric reduction can hide a particle-hole-asymmetric instability. A coherent fSWAP angle error $\theta = \pi/2 + \delta\theta$ should be tested

separately from classical ancilla preparation error.

Wall separation and aspect ratio. Vary both separations on the torus, not only N_y . Test the scaling variable in Eq. (D25) using an independently measured $\xi_{\perp}$, and compare against a domain-wall-off control at the same dimensions and observation time.

Tangent numerics. Report burn-in and observation windows, QR cadence, the number of propagated vectors, singular tolerance, null counts, and censored trajectories. Compare the direct chronological product at small size with QR accumulation. At every fixed finite size, extend or extrapolate the QR observation time before fitting any size dependence. Randomize the initial tangent frame and localize the projector onto the complete near-marginal subspace rather than one basis-dependent QR vector. For a half-strip, count $N_{=, \neq}^{\theta}$ and verify Eq. (D7) from a pure product input.

Fit and sampling stability. For entropy, charge fluctuations, and correlations, vary the fit window and include the full trajectory bootstrap. Report the number of independent Born trajectories at every size. Heatmaps are secondary localization diagnostics and should not substitute for integrated wall weight, inverse participation ratios, or transverse localization-length fits.

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Robustness and control matrix

Proposed panels: finite-support convergence; product, generic-Gaussian, and maximally mixed initial states; raster versus finite-color schedules; feedback and ancilla errors; wall-separation scaling; and null/censor statistics for the tangent QR. This is a placeholder layout only. The final figure should display all actual run parameters and should not combine data from microscopically different domain-wall constructions without labels.

FIG. 11. Robustness and control matrix. This is a layout placeholder; the production acceptance criteria and complete estimator definitions are given in the text.

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